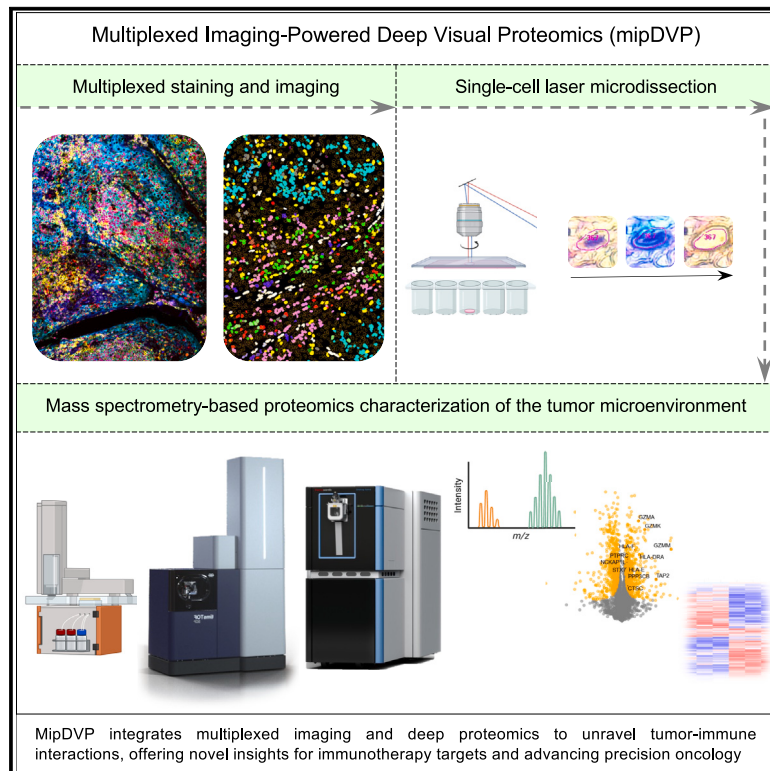


Deciphering functional tumor-immune crosstalk through highly multiplexed imaging and deep visual proteomics

Graphical abstract



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In brief

Xiang Zheng et al. develop an advanced approach combining multiplexed imaging and deep visual proteomics to spatially map the tumor microenvironment. By analyzing colorectal and tonsil cancer tissues, they uncover immune-modulating macrophage barriers and T cell adaptations in hypoxic regions. These findings provide critical insights into immune interactions and potential therapeutic targets, advancing precision oncology.

Highlights

- Multiplexed imaging and deep visual proteomics of the tumor microenvironment
- Identified 21 cell populations and thousands of proteins in a spatial tissue context
- Revealed macrophage barriers to T cell infiltration and hypoxic T cell adaptations
- Provided spatial insights into immunotherapy targets for personalized oncology



Resource

Deciphering functional tumor-immune crosstalk through highly multiplexed imaging and deep visual proteomics

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SUMMARY

Deciphering the intricate tumor-immune interactions within the microenvironment is crucial for advancing cancer immunotherapy. Here, we introduce mipDVP, an advanced approach integrating highly multiplexed imaging, single-cell laser microdissection, and sensitive mass spectrometry to spatially profile the proteomes of distinct cell populations in a human colorectal and tonsil cancer with high sensitivity. In a colorectal tumor—a representative cold tumor—we uncovered spatial compartmentalization of an immunosuppressive macrophage barrier that potentially impedes T cell infiltration. Spatial proteomic analysis revealed distinct functional states of T cells in different tumor compartments. In a tonsil cancer sample—a hot tumor—we identified significant proteomic heterogeneity among cells influenced by proximity to cytotoxic T cell subtypes. T cells in the tumor parenchyma exhibit metabolic adaptations to hypoxic regions. Our spatially resolved, highly multiplexed strategy deciphers the complex cellular interplay within the tumor microenvironment, offering valuable insights for identifying immunotherapy targets and predictive signatures.

INTRODUCTION

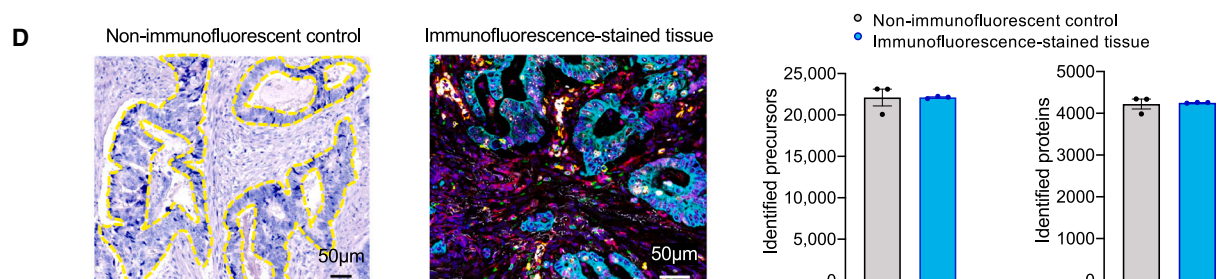
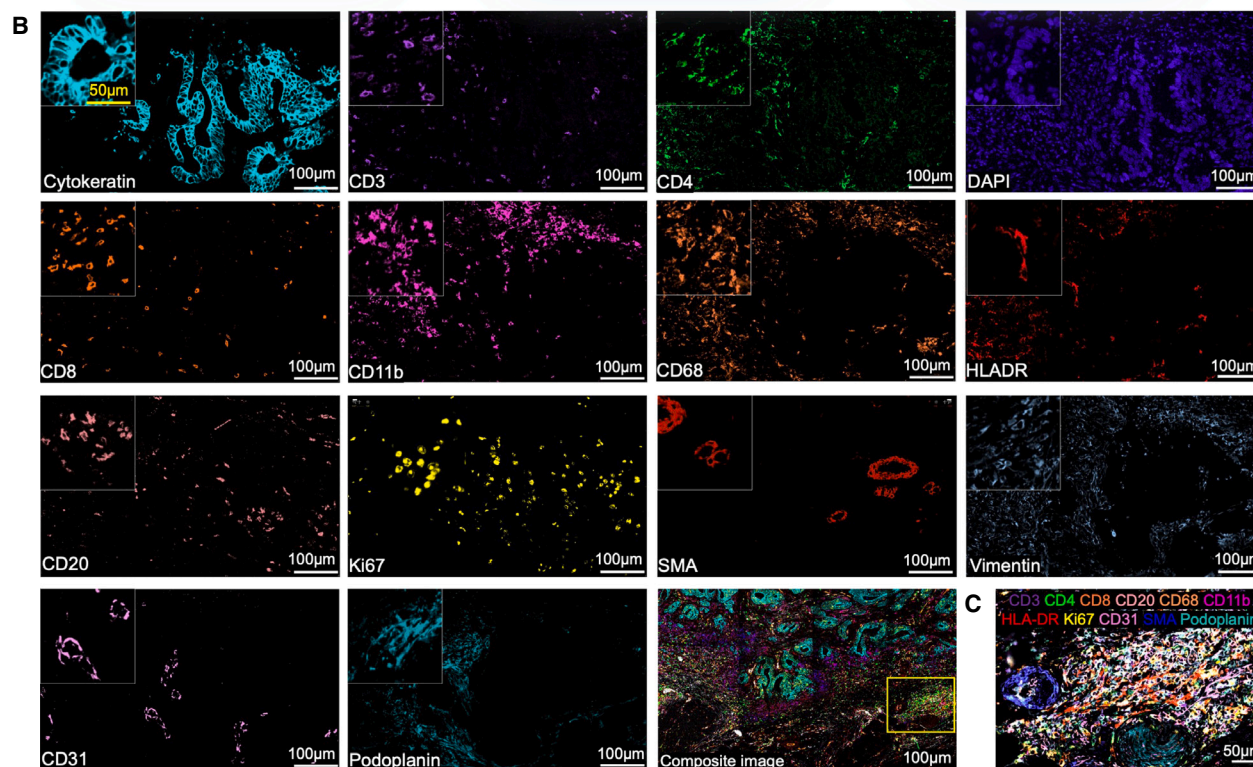
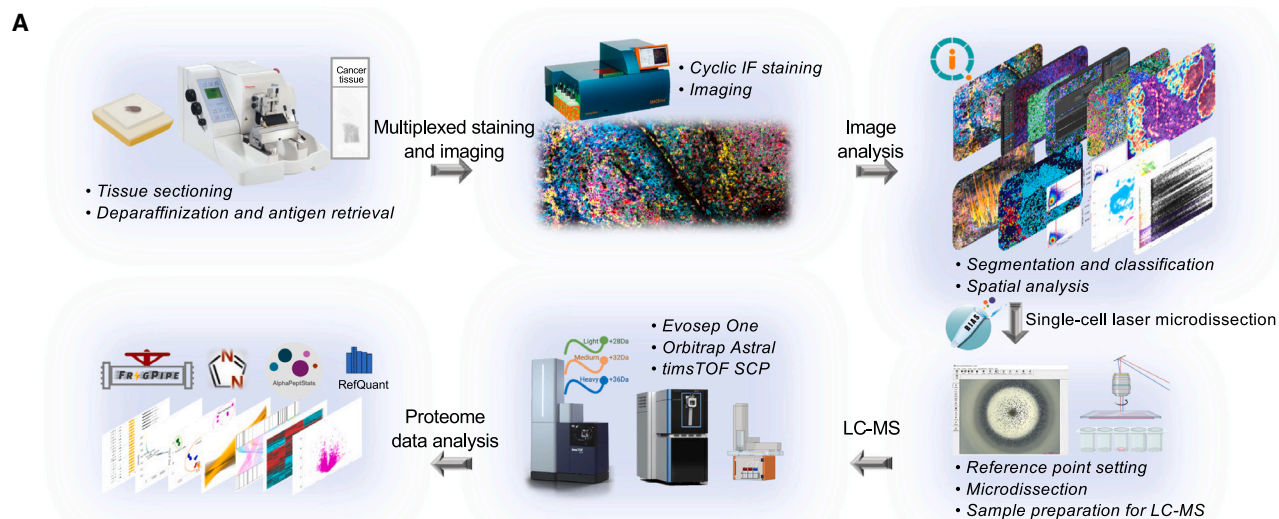
The tumor microenvironment (TME) is a complex ecosystem where cancer cells dynamically interact with diverse non-malignant cells, including immune cells, fibroblasts, and endothelial cells, as well as the extracellular matrix.¹ These intricate interactions shape cancer progression and therapy response.² In particular, crosstalk between cancer cells and immune cells within the TME plays a central role in modulating anti-tumor immunity, often leading to immunosuppression, which enables tumor immune evasion.³ Deciphering the cellular composition, spatial organization, and molecular circuitry of the TME is therefore crucial for understanding tumor biology and developing effective immunotherapies.

Advances in spatial omics technologies have revolutionized our ability to dissect tumor heterogeneity.^{4,5} Unbiased spatial transcriptomics methods, such as 10X Genomics Visium and Stereo-seq, leverage deep sequencing technologies to provide high-resolution spatial profiling without prior gene selection.⁶ These innovations enable comprehensive mapping of gene expression patterns within the TME.⁷ At the protein level, conventional single or low-plex staining methods often fall short in retaining spatial relationships across multiple cell populations,

and they do not achieve the highly multiplexed characterization needed to capture the full complexity of the TME.^{7,8} Although imaging-based approaches like multiplexed ion beam imaging and imaging mass cytometry can spatially profile targeted proteins, their limited depth hinders unbiased discovery of the cellular networks within the TME.^{7,9}

Mass spectrometry (MS)-based proteomics has long been used to characterize tumor tissue at the bulk level, with notable success, with the Clinical Proteomic Tumor Analysis Consortium (CPTAC) as a prominent example.^{10–12} For instance, CPTAC recently released a pan-cancer proteogenomic dataset enabling multi-omics analysis of bulk tumor samples.^{10,13} Nevertheless, despite tremendous technological improvement in MS-based proteomics,¹⁴ these datasets lack the spatial resolution needed to dissect the complex cellular interplay within the heterogeneous TME at the single-cell-type level. Achieving highly multiplexed characterization of cells together with deep single-cell-type proteome characterization requires innovative analytical strategies. We previously introduced deep visual proteomics (DVP), a spatial proteomics technology integrating imaging, cell segmentation and classification, laser microdissection, and MS to investigate complex tissues.¹⁵ However, DVP has only been demonstrated with up to four-plex staining so far.





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Here, we aim to further develop multiplexed-imaging-powered DVP (mipDVP), an advanced approach that integrates highly multiplexed imaging with MS multiplexing through a recently developed data-independent acquisition (mDIA) workflow.^{16,17} By optimizing multiplexed staining to maintain morphological integrity, we achieved precise single-cell laser microdissection and deep spatial proteomics of lymphocytes, overcoming challenges posed by their small size to provide insights into their complex biology in the native tissue microenvironment.¹⁸ We evaluated our method's versatility by applying it to both immunologically cold and hot tumors—those less and more likely to respond to immunotherapy, respectively¹⁹—demonstrating its wide applicability across diverse TME. Using this integrated approach, we uncovered spatial compartmentalization of immunosuppressive macrophages and T cells in a colorectal tumor (a “cold” tumor case), revealing distinct functional states of T cells in different tumor compartments. Conversely, in a tonsil cancer sample (a “hot” tumor case), we identified significant proteomic heterogeneity among cells influenced by proximity to cytotoxic T cell subtypes. Notably, T cells in tumor parenchyma exhibited metabolic adaptations to hypoxic regions. By applying mipDVP, we aim to decipher the complex cellular interplay and functional adaptations within the TME, potentially discovering immunotherapy targets. More broadly, our framework provides a powerful means for ultra-high sensitivity profiling of limited clinical samples to inform personalized oncology.

RESULTS

Integrated 14- to 22-plex imaging and MS for single-cell-type tumor spatial proteomes

To characterize the spatial molecular landscape of the TME at the single-cell level, we first performed 14- to 22-plex immunofluorescence staining on formalin-fixed paraffin-embedded (FFPE) sections from a human colorectal cancer (CRC) sample and a tonsil cancer sample using the MACSima imaging platform (Figures 1A–1C, S1A–S1C, and S2A; STAR Methods and see below). For 22-plex staining, this took less than 50 h on a tissue region of 30 mm², sufficient to isolate even rare cell types from the same tissue slide for downstream analysis.

Unlike conventional low-plex staining, multiplexed imaging revealed a plethora of diverse cell types and their spatial interactions within the same tissue section (Figures 1B and 1C). Multiplexed staining of the CRC tissue enabled detailed evaluation of the spatial distribution of tumor, immune, and stromal cells. Tertiary lymphoid structures (TLS) are important sites of local immune responses within the TME.²⁰ Within the lamina propria—a layer beneath the tumor epithelium—we discerned TLS composed of CD3+CD4+CD8– helper T cells (TH), CD3+CD8+CD4– cytotoxic T cells (CTLs), CD20+CD3– B cells, and

CD68+ macrophages. This region also contained CD31+ blood vessels and Podoplanin+ lymphatic vessels (Figure 1C).

Next, we utilized high-plex imaging data to perform cell segmentation and classification,²¹ which was essential for the subsequent laser microdissection and MS-based proteomics analysis²¹ (STAR Methods).¹⁵ Cell shapes of the same type and region were collected in 384-wells followed by liquid chromatography-mass spectrometry (LC-MS) analysis using very low flow Whisper gradients (100 nL/min).²² For ultimate sensitivity, we used both diaPASEF on the timsTOF and the Astral platforms.^{23–25} All MS data were analyzed using DIA-NN.²⁶ Our workflow analyzed 40 samples per day using a label-free MS method and 80 samples with the dimethyl-labeling method,¹⁶ for a total of 13–26 cell types with technical triplicates per day.

To minimize potential proteomic alterations due to the cyclic multiplexed imaging process, which involves multiple rounds of staining and imaging, we used photobleaching instead of chemical stripping. The MACSima platform facilitated direct and automated antibody pipetting onto membrane slides within a hub, reducing slide handling between imaging cycles. To evaluate the impact of MACSima multiplexed staining on proteome integrity, we isolated tissue areas of 40,000 μm², equivalent to approximately 60 CRC cells, from both immunofluorescence-unstained control and MACSima-stained CRC tissues (Figures 1D and S2B). Comparative analysis of the precursor and identified protein profiles revealed only minimal protein loss during the staining process (no loss of protein identifications after 14-plex), confirming the compatibility of MACSima imaging with downstream laser microdissection and the efficiency of protein recovery for MS-based proteomics (Figure 1D). Similarly, we observed minimal protein loss during the staining process for the tonsil cancer tissue (less than 2% loss of protein identifications after 22-plex staining) (Figures S2C and S2D), demonstrating the broad applicability of this approach across different tumor types.

Macrophages establish an immunosuppressive barrier, potentially hindering T cell infiltration in a colorectal cancer case

To investigate the TME with respect to immunologically hot or cold properties, we conducted 14-plex staining on a CRC tumor (STAR Methods). We identified distinct cell populations within this CRC tissue, including tumor cells, macrophages, CTLs, TH, and B cells, as well as blood vessels and lymphatic vessels (Figures 1B and 2A). The presence and spatial distribution of these cell types are crucial for understanding the immunological landscape of the tumor, which, in turn, informs the choice of immunotherapy.

The image analysis yielded two main findings: first, we observed a remarkable accumulation of immune cells within

Figure 1. Integrated 14- to 22-plex imaging and MS for single-cell-type tumor spatial proteomes

(A) Schematic of the mipDVP workflow integrating multiplexed staining and mass-spectrometry-based proteomics to enable deep proteomic profiling while preserving the spatial context of the tissue microenvironment.
(B) Multiplexed pseudo-colored images of marker staining of colorectal cancer FFPE sections.
(C) Representative image of the tertiary lymphoid structure outlined in yellow in (B).
(D) Comparison of the number of precursor and protein identifications of tumor epithelial regions from immunofluorescence-unstained and MACSima-stained tissues. The yellow dotted line indicates the boundary of the tumor epithelial region, while the pseudocolor cyan represents CK+ tumor cells ($n = 3$).

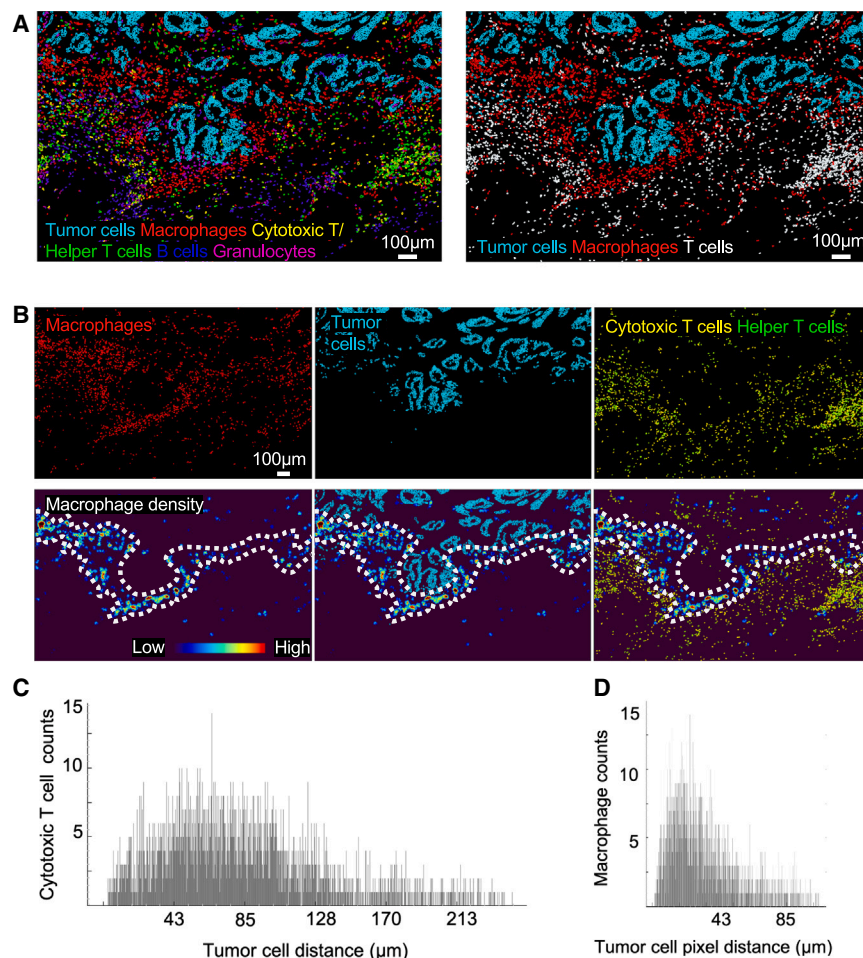


Figure 2. Macrophages establish an immunosuppressive barrier potentially hindering T cell infiltration in a colorectal cancer case

(A) Phenotype map showing the spatial distribution of the indicated cell types.

(B) Representative image of a macrophage-abundant region with overlaid masks highlighting tumor cells and T cells.

(C–D) Histogram of the counts of cytotoxic T cells (C) or macrophages (D) as a function of their proximity from tumor cells in colorectal cancer case.

the lamina propria, positioned beneath the tumor epithelium (Figure 2A). Second, we noted a barrier-like positioning of macrophages between the tumor epithelium and lamina propria, with a low expression of HLA-DR (Figures 2B and S2E), suggesting their reduced antigen-presenting capacity and involvement in immunosuppression.²⁷ Additionally, we observed a higher abundance of immune cells, including CTLs and TH, infiltrating the lamina propria compared with the tumor epithelial region (Figure 2B).

Spatial distribution analysis revealed the proximity of macrophages (peak count at distance of 20 µm) to the tumor epithelium compared with CTLs (peak count at distance of 68 µm). This indicates that macrophages tend to be located closer to the tumor regions than T cells (Figures 2C and 2D). This pattern suggests that macrophages might play a role in shaping the immune landscape.²⁸ To functionally characterize macrophages in a spatial context and address their potential immunosuppressive roles, we performed single-cell microdissection of macrophages from both tumor epithelial and lamina propria regions, followed by proteome profiling. Our analysis quantified over 4,000 protein groups from macrophages derived from the equivalent of 100 cells (STAR Methods), with more than 1,000 proteins significantly differentially expressed between macrophages from the

lamina propria and those from the tumor epithelial region (Figure S2F). Notably, macrophages at the tumor epithelial region exhibited a downregulation of pathways associated with macrophage activation, interferon-gamma response, and phagocytic capability (Figure S2G). Key molecules, including AIF1, ATM, C1QA, CD74, CLU, ITGAM, ITGB2, MYO18A, MYO1C, PLCG2, PTPRC, and SYK, which are crucial for macrophage activation, were significantly downregulated in macrophages at the tumor epithelial region, with fold changes ranging from 1.3 to 4 (Figure S2F, S2G, and S3A). Among these, CD74, ITGAM, PTPRC, SYK, and AIF1 are specifically associated with anti-tumoral M1 macrophage activation, suggesting a reduced M1-like phenotype of macrophages in the tumor epithelial region.^{29–32} This is consistent with the

observed low HLA-DR expression and further supports the notion of a less immunostimulatory and potentially immunosuppressive macrophage population in this region. Conversely, these macrophages showed upregulation of pathways related to the citric acid cycle and precursor protein transport and processing, indicating a shift toward a more metabolically active state, which is characteristic of a tumor-supporting M2-like phenotype³³ (Figure S2G). These proteomic findings provide an in-depth characterization of the macrophages, suggesting an enhanced immunosuppressive phenotype of macrophages in the tumor epithelial region.

The presence of at least a few T cells in the tumor epithelial region further classifies it as “T cell excluded” rather than “immune desert.”³⁴ The distinct infiltration pattern of T cells and macrophages in the lamina propria and tumor epithelium, separated by a barrier formed by macrophages, suggested the need for future proteomic analysis of both cell types in these regions to elucidate functional differences between immune cell subsets.

T cell subsets have distinctive spatial proteomic signatures in colorectal TME

To functionally characterize T cells in a spatial context, we conducted single-cell microdissection of CTLs and TH from both

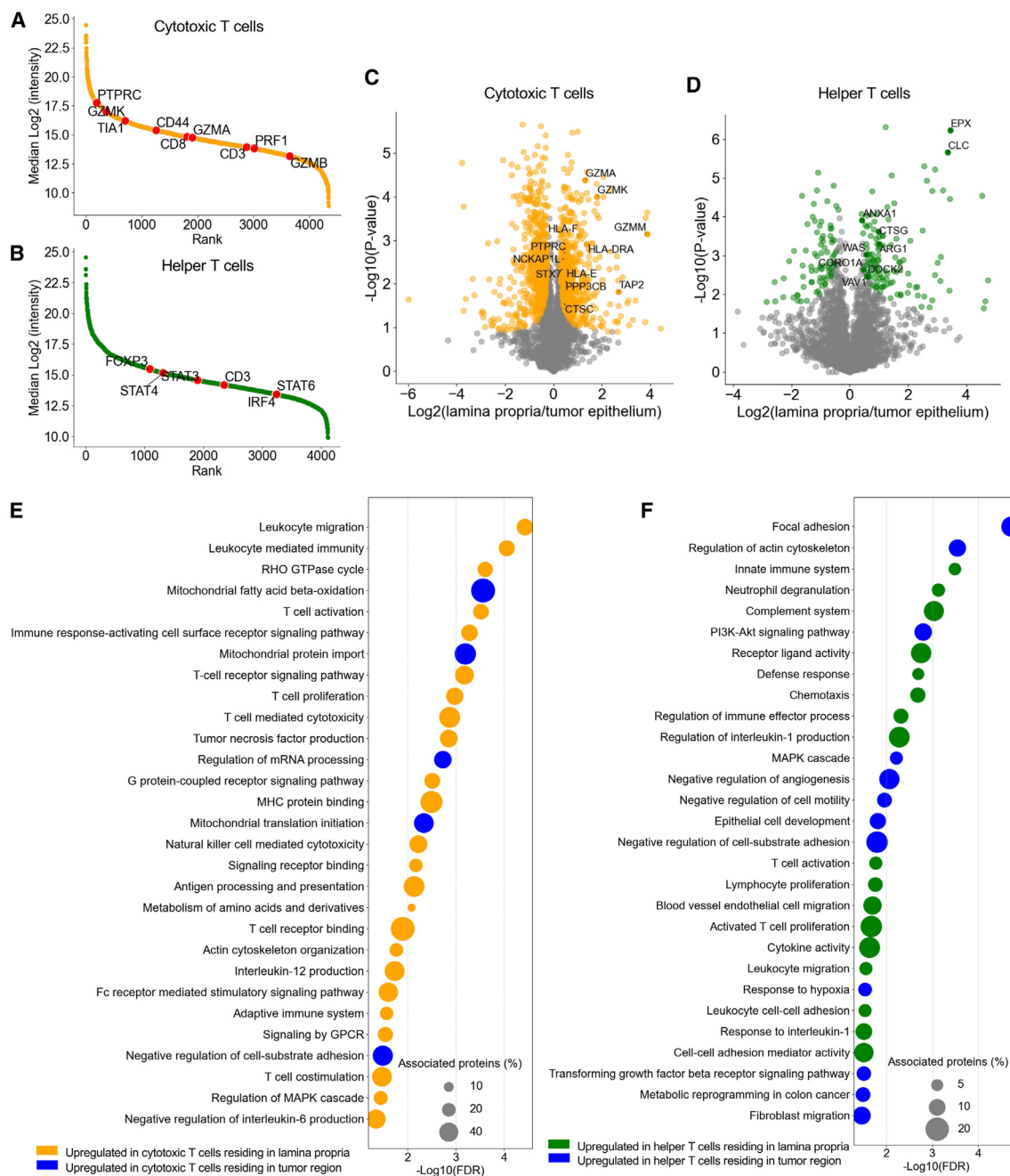


Figure 3. T cell subsets have distinctive spatial proteome signatures in colorectal tumor microenvironment

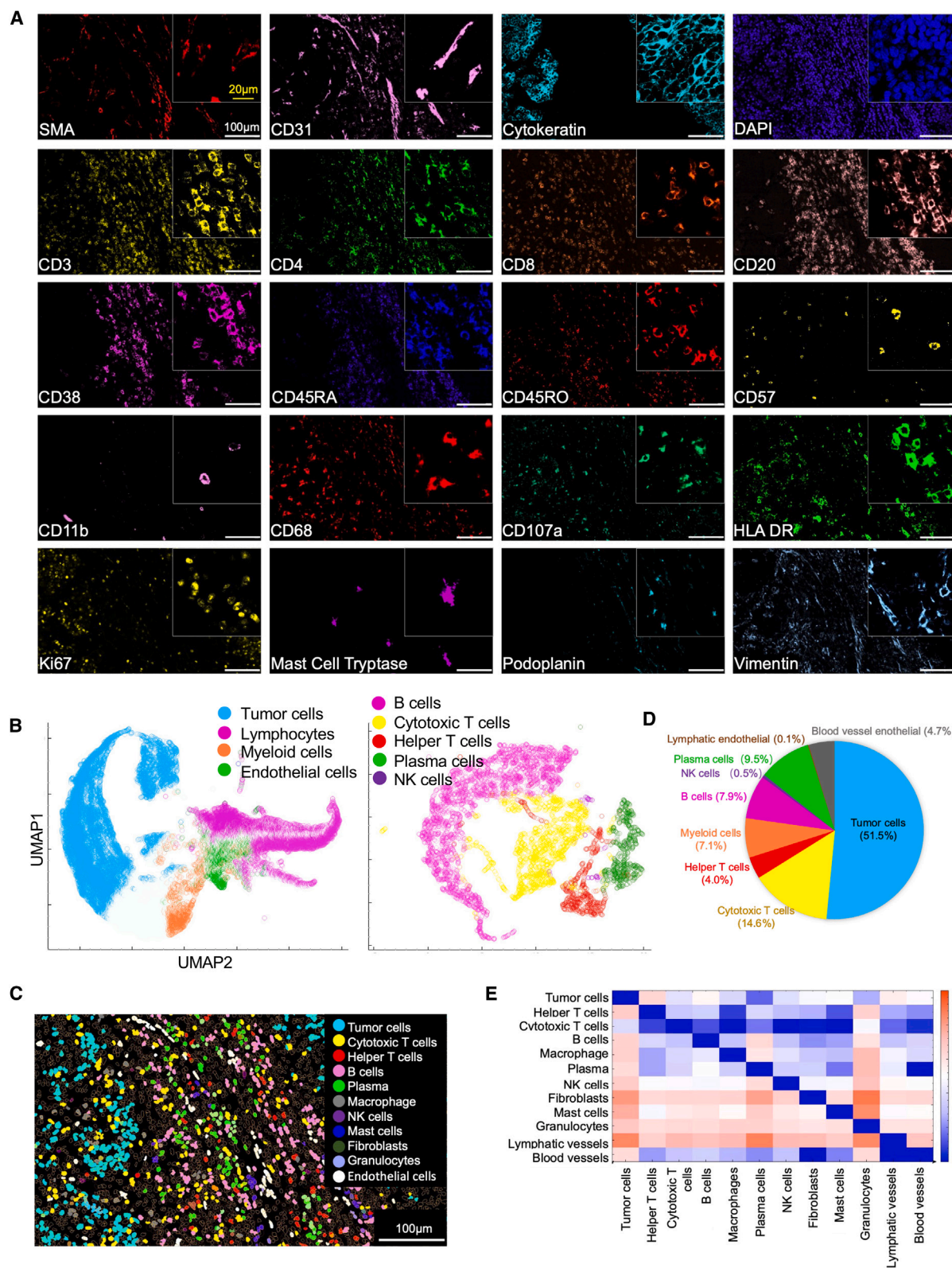
(A and B) Ranking of protein quantified in cytotoxic T cells (CTLs) (A) and helper T cells (TH) (B), ranked by median transformed intensity values detected by mass spectrometry, with cell-type-specific markers highlighted.

(C and D) Volcano plots illustrating proteomic comparisons of CTLs (C) and TH (D) between the lamina propria and tumor epithelial regions, with T cell-activity-related markers highlighted. Significantly enriched proteins are denoted as yellow or green dots (two-sided t test, false discovery rate [FDR] < 0.05, S0 = 0.1, n = 3).

(E and F) Gene Ontology (GO) term analysis of significantly altered proteins in CTLs (E) and TH (F) residing in the lamina propria and tumor epithelial regions.

tumor epithelial and lamina propria regions, followed by deep proteome profiling. Our analysis quantified more than 4,000 protein groups with coefficients of variation between 8% and 18% in both CTLs and TH from the equivalent of only 200 of these cells

(STAR Methods; Figures 3A, 3B, S3B, and S3C). This revealed substantial differences in protein expression profiles for the same cell types between the two regions. Interestingly, unbiased proteomics revealed the expression levels of markers of CTLs



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(including CD3, CD8, perforins, and granzymes) and TH (including CD3, FOXP3, and IRF4). Overall, more than 1,000 proteins were significantly differentially expressed proteins in CTLs between the lamina propria and tumor epithelial regions (Figure 3C) and about 200 in TH between the same regions (Figure 3D).

Comparative analysis of the quantitative proteomes of CTLs revealed distinct functional characteristics between those residing in the lamina propria and tumor epithelium. Pathways associated with T cell activation, cytokine production, immune receptor signaling, and antigen processing and presentation were downregulated in CTLs residing in the tumor epithelial region, indicating reduced immune response against cancer cells (Figure 3E). By contrast, in the lamina propria, pathways related to cytotoxicity, such as natural-killer-cell-mediated cytotoxicity, T cell-mediated cytotoxicity, major histocompatibility complex (MHC) protein binding, tumor necrosis factor production, and leukocyte-mediated immunity, were elevated, suggesting increased effectiveness in direct immune response. These CTLs also had a comparatively upregulated mitogen-activated protein kinase (MAPK) cascade, GPCR signaling, interleukin-12 production, and T cell receptor signaling pathway, implying enhanced cellular communication and activation of intracellular signaling cascades crucial for immune response regulation. Their effective functioning appears to be supported by elevated metabolic pathways. Moreover, upregulation of cell migration and adhesion suggests active movement within tissues to locate and engage with cancer cells. Conversely, CTLs in the tumor epithelium exhibited upregulated mitochondrial metabolism, including mitochondrial fatty acid beta-oxidation, to adapt to the TME. Additionally, MS-based proteomics pinpoints the identity and fold-change of the involved molecular players. For instance, molecules including CTSC, GZMM, GZMA, GZMK, HLA-DRA, HLA-E, HLA-F, NCKAP1L, PPP3CB, PTPRC, STX7, and TAP2, which play crucial roles in T cell cytotoxicity, exhibited significant downregulation in CTLs located in the tumor epithelium, with fold changes ranging between 2 and 16 (Figures 3C and S3D).

Similarly, the proteomics analysis of TH in the tumor epithelial region revealed distinct functional roles compared with those in the lamina propria. In the tumor epithelium pathways associated with adaptation to hypoxic TME, metabolic reprogramming, negative regulation of angiogenesis, and cell adhesion were upregulated, suggesting their involvement in modulating immune responses, tumor progression, and interactions with the TME (Figure 3F). Furthermore, growth factor signaling, such as the transforming growth factor beta receptor signaling pathway, and intracellular signaling cascades, including phosphatidylinositol 3-kinase (PI3K)-Akt signaling and MAPK cascade, indicated alterations in responsiveness to external

stimuli. These changes may enable TH in the tumor epithelium to actively participate in tumor cell proliferation, survival, and immune evasion mechanisms. Upregulated pathways in lamina propria-residing TH included T cell activation, interleukin-1 response, and monocyte chemotaxis. This suggests increased antigen processing and presentation capability and increased immune surveillance. The activation of pathways linked to the innate immune system, such as modulation of the complement system and neutrophil degranulation, alongside elevated cell-cell adhesion, underscores the significant role of lamina-propria-residing TH in modulating immune responses, immune cell localization, and communication. Molecules including ANXA1, ARG1, CLC, CORO1A, CTSG, DOCK2, EPX, VAV1, and WAS, which play crucial roles in T cell activation, exhibit significant downregulation in TH located in the tumor epithelium, with fold changes ranging between 1.5 and 16 (Figures 3D and S3E). In the CRC case, IF analysis of CD4⁺ TH cells across the entire tissue section served as a cross-validation of our MS-based proteomics data. By quantifying the expression levels of specific markers throughout the whole tissue, we confirmed the consistent upregulation of proteins such as ANXA1, ARG1, CLC, and EPX in TH cells, aligning with our proteomic findings (Figure S4).

These findings highlight the distinct functional roles of CTLs and TH residing in different microenvironments within this CRC case, directly at the level of the proteins involved. Our data provide detailed insights into the interactions with the TME, with potential implications for tumor progression and immune response modulation.

Cytotoxic T cells are in close proximity to cancer cells in a tonsil cancer case

To evaluate the versatility of mipDVP, we additionally extended it to a tonsil cancer FFPE section, incorporating expanded marker staining (22-plex) to enhance the detection of a more diverse range of cell types (Figures 4A, S2A, and S2C). Given our extensive range of markers, we were able to map intercellular spatial relationships between cell types, enabling investigation of the interplay between cells in the TME. We assessed the expression of cytokeratin (CK), CD3, CD4, CD8, CD20, CD38, CD57, CD11b, CD68, CD31, Podoplanin, HLA-DR, and Ki67, based on the staining, and performed uniform manifold approximation and projection of 57,276 cells. We identified four primary cell clusters: tumor cells, lymphocytes, myeloid cells, and endothelial cells. In-depth examination of lymphocytes further revealed distinct populations, comprising CTLs, TH, B cells, plasma cells, and natural killer (NK) cells (Figures 4B and 4C). Notably, immune cells constituted about 43% of the total identified population in the tumor (Figure 4D), highlighting substantial immune cell presence.

Figure 4. Cytotoxic T cells are in close proximity to cancer cells in a tonsil cancer case

- (A) Multiplexed staining on a tonsil cancer tissue section. Scale bar (in white), 100 μ m.
 (B) UMAP revealing four distinct cellular clusters. The lymphocyte cluster further divides into five subclusters, based on the expression of CK, CD3, CD4, CD8, CD20, CD38, CD57, CD11b, CD68, CD31, Podoplanin, HLA-DR, and Ki67.
 (C) Colored masks represent cellular classification in a representative region.
 (D) Tumor cells constitute about half of classified cell types, with immune cells as the second largest category (43%), whereas the remainder are stromal cells.
 (E) Distance heatmap depicting the proximity and interactions among classified cell types, with lower values indicating higher proximity (STAR Methods).

To gain insights into the spatial organization of the TME, we evaluated the spatial distances among diverse cellular populations. To do so, we determined all pairwise pixel distances between different cell types, averaged and normalized them, and constructed a heatmap (STAR Methods; Figure 4E). This analysis revealed that CTLs were positioned closely to tumor cells, with an average normalized distance value of 6 (Arcsinh normalized and averaged pixel distances among individual cells, ranging from 4 to 9, indicating closer to farther distance, respectively). In comparison, the distance values within the same cell type were the closest at 4, while distances between tumor cells and other cell types, such as TH, B cells, macrophages, and NK cells, ranged from 7 to 9, indicating greater distance. Furthermore, CTLs were found to be relatively close to other immune and stromal cells, including TH, B cells, macrophages, plasma cells, NK cells, endothelial cells, and fibroblasts, with average normalized distances ranging from 5.5 to 6.5. This proximity suggests significant presence of CTLs in both the tumor parenchyma and stroma, potentially facilitating interactions or cross-talk between CTLs and tumor cells, indicating that this TME is immunologically hot.¹⁹

The close positioning of CTLs to tumor cells in this tonsil cancer case, as revealed by our spatial analysis, reveals where direct tumor-immune interactions and anti-tumor immune responses take place. The abundant immune cell presence and close proximity of CTLs to tumor cells suggest a more permissive TME for anti-tumor immunity, which may contribute to the immunologically hot nature of this cancer type. This physical proximity is in stark contrast to the spatial compartmentalization of immunosuppressive macrophages and T cells in the CRC case. These insights highlight the importance of considering spatial context and cellular interactions when assessing the immunological status of a tumor and its potential responsiveness to immunotherapies.

CTLs exhibit close proximity to tumors, with a suppressive impact on tumor proliferation

We found that CTLs not only abound in the stroma but also proximal to the tumor parenchyma, whereas TH primarily reside in the stroma (Figure 5A). This heightened proximity of CTLs to tumor cells as compared with TH indicates their potential role in tumor surveillance and anti-tumor response.

CD45RA and CD45RO, transmembrane glycoproteins found on leukocytes, are markers for resting T cells and memory T cells activated by antigens, respectively.³⁵ Our analysis identified two primary subpopulations: CD45RA⁺CD45RO[−] (abbreviated as RA⁺RO[−]) as the most prevalent, followed by CD45RA⁺CD45RO⁺ (abbreviated as RA⁺RO⁺) (Figures 5B and 5C). Notably, both CTLs and TH within the RA⁺RO[−] subgroup exhibited a lack of PD1 expression (Figure 5C).

In the CTLs in tumor parenchyma (TIL-CTLs), we unexpectedly found a new phenotype, namely TIL-CTLs that had lost expression of CD45RA and CD45RO (Figures 5B and 5D). The observed diminished CD45RA or CD45RO expression in TIL-CTLs suggests a specialized functional adaptation within the TME, warranting further investigation (Figure 5D). Among all CD45RA and CD45RO-defined lymphocyte subpopulations, RA⁺RO[−] CTLs were the closest type in proximity to

tumor cells, further supporting the notion that TIL-CTLs are RA⁺RO[−] (Figure 5E).

To assess the functional impact of TIL-CTLs on tumor cells, we investigated the relationship between TIL-CTL proximity and tumor cell proliferation. Remarkably, we observed a suppressive effect of TIL-CTLs on tumor cell proliferation, as evidenced by a higher ratio of Ki67[−]/Ki67⁺ tumor cells (indicating reduced proliferation) in tumor cells located within close proximity (<30 μ m) to TIL-CTLs compared with those at intermediate distances (>30 μ m and <60 μ m) (Figure 5F). This finding aligns with the observed upregulation of the degranulation marker CD107 on CTLs, a critical factor for enhanced cytotoxic activity³⁶ (Figure S5A). Unexpectedly, tumor cells located more than 90 μ m away from TIL-CTLs exhibited a significant increase in the Ki67[−]/Ki67⁺ ratio (Figure 5F). This suggested heightened immune surveillance in regions where tumor cells are more distant from TIL-CTLs and closer to the stroma, with potential implications for regulating tumor growth.

To further investigate this unexpected phenomenon and gain insights into its underlying molecular mechanisms, we performed spatial and cell-type-specific unbiased proteomics. To this end, we isolated tumor cells according to their proximity to TIL-CTLs, categorizing them as close, intermediate, and far for subsequent proteomic analysis. The premise of our strategy was that the proteomic diversity of tumor cells, based on their distance from TIL-CTLs, might offer insights into the molecular landscape and potential functional consequences of these distinct tumor cell populations (Figure 5G).

Proteomic heterogeneity in tumor cells correlates with proximity to TIL-CTLs

Tumor cells located at various proximities—close (<30 μ m), intermediate (>30 μ m and <60 μ m), and distant (>60 μ m and <90 μ m)—to TIL-CTLs exhibited different proteomic signatures (Figures 5G, 6A and 6B, and S5B–S5D). Principal-component analysis (PCA) revealed distinct grouping of these tumor cells according to their proximity to TIL-CTLs, with those in the distant group clustering notably apart from the close and intermediate groups (Figure 6B).

The upregulated interferon and interleukin-12 signaling in tumor cells close to TIL-CTLs indicate that tumor cells respond to cytotoxic activity of CTLs and Th1 (Figure 6C). The key players involved in this include JAK1, STAT1, IRF9, HPX, and CRKL (Figure 6D). Conversely, tumor cells distant from TIL-CTLs and in proximity to stroma exhibited a higher Ki67[−]/Ki67⁺ tumor cell ratio compared with those in closer proximity (Figure 5F), which was corroborated by increased pathway activity in complement activation, immunoglobulin-mediated immune response, and phagocytosis engulfment in these distant tumor cells (Figure 6C). This suggests active immune surveillance, potentially from macrophages, B cells, and plasma cells in the stroma. Proteins involved in this response, including C3, C4A, and ITGA2, are known to increase the interaction between macrophages and tumor cells, facilitating phagocytosis³⁷ (Figure 6D). The high density of macrophages, B cells, and plasma cells in the tumor stroma may indicate physical interactions between them and tumor cells at the stromal edge (Figure 6E). Blood vessels were predominantly present within the tumor stroma, making the

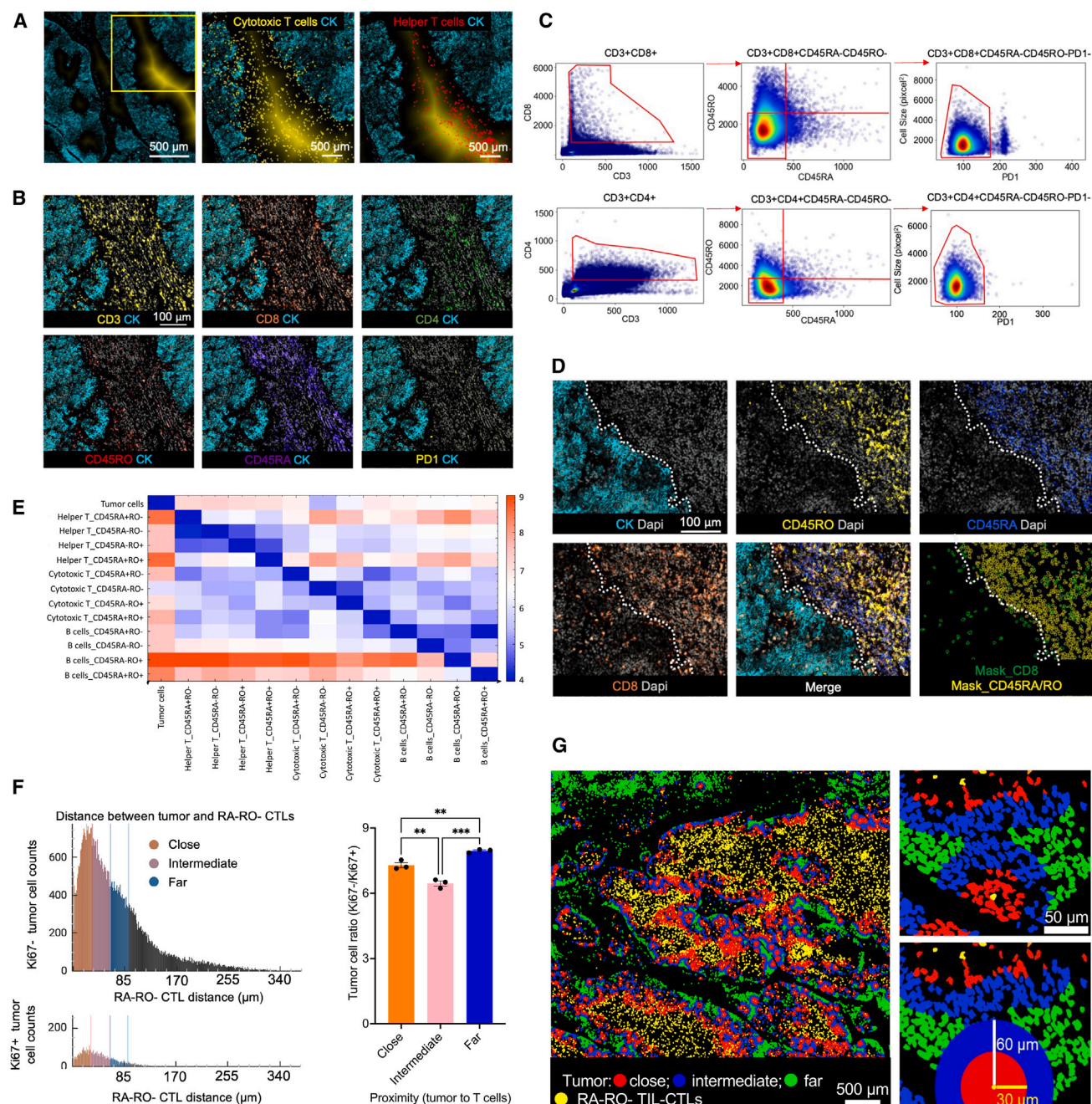


Figure 5. Cytotoxic T cells (CTLs) exhibit close proximity to tumors with a suppressive impact on tumor proliferation

(A) Density gradient in the background indicates distance to tumor islands, with brighter yellow representing greater distance from tumor cells. Left panel only has the gradient whereas the middle, zoomed panel additionally depicts yellow masks of the CTLs and the right panel red dots of helper T cells (TH), respectively.

(B) Representative images showcasing markers for T cell phenotyping.

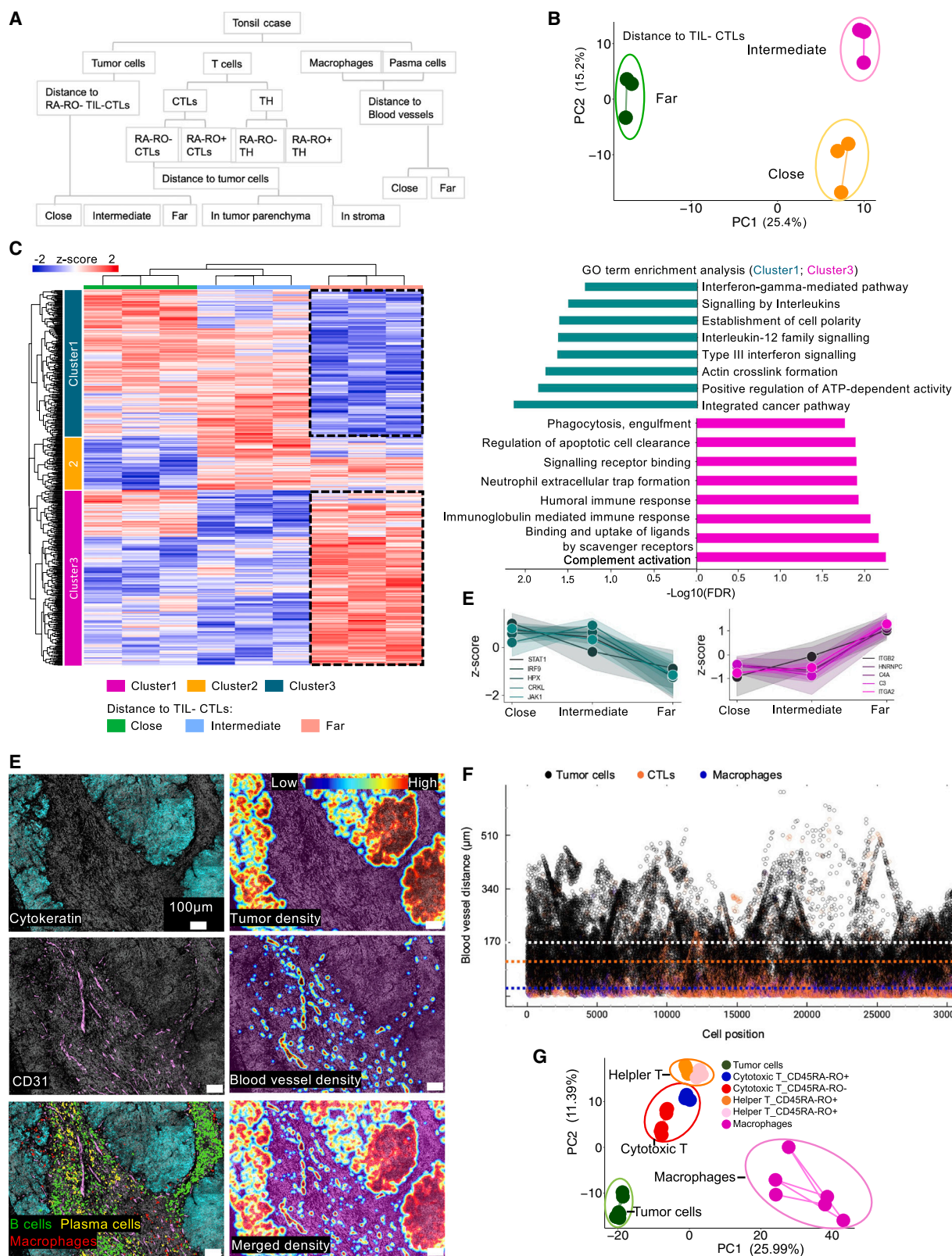
(C) T cell gating based on the expression of CD3, CD8, CD45RA, CD45RO, and PD1 to analyze the corresponding T cell subpopulations.

(D) Representative images showing CD3- and CD8-positive CTLs in tumor parenchyma (TIL-CTLs) with absent CD45RA and CD45RO expression.

(E) Distance heatmap depicting proximity and interactions between tumor cells and T and B lymphocyte subtypes, with lower values indicating higher proximity as in Figure 4E.

(F) Histogram of Ki67-/Ki67+ tumor cell number as a function of distance from RA-RO- CTLs. Data represent three technical replicates. Statistical significance was determined using one-way ANOVA. *** $p < 0.001$, ** $p < 0.01$.

(G) Representative image displaying masks of tumor cells at varying proximities to RA-RO- TIL-CTLs.



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distance of various cell types to these vessels a reliable indicator of their closeness to the tumor stroma itself (Figure 6E).

To assess the spatial distribution across the entire imaged area and to present this information through dimensionality reduction techniques, we charted the x axis positions of tumor cells, macrophages, and CTLs relative to their pixel distance from blood vessels. A higher pixel distance value indicates a greater distance of tumor cells, macrophages, and CTLs from the blood vessels located in the stroma. The median distance from blood vessels to tumor cells was 170 μm , whereas this was 119 μm to CTLs and 17 μm to macrophages, suggesting that while vascular supply supports overall tumor growth, TIL-CTL proximity is more directly linked to proliferation suppression. Thus, based on physical distance, tumor cells far from blood vessels could be attacked by CTLs, while those closer to vessels also by macrophages (Figures 6E and 6F).

To investigate this on a molecular basis, we isolated macrophages from areas close and far from blood vessels for proteomic analysis (Figure 6F). We set a cutoff at 85 μm to distinguish between close and far distances to blood vessels, based on it being half of the median distance observed between tumor cells and blood vessels (Figure 6F). PCA analysis revealed distinct grouping of tumor cells, T cells, and macrophages (Figure 6G). The notable upregulation of membrane dynamics regulators, such as ACAP1 in macrophages near blood vessels, suggests increased cellular mobility, aiding their migration from blood vessels to tumor sites³⁸ (Figure S5E). These findings, consistent with image analysis, underscore the functional interplay and cellular dynamics between the immune system and tumor cells within the tumor parenchyma and its immune-enriched stroma.

TIL-CTLs exhibit strategic adaptations to hypoxic TME

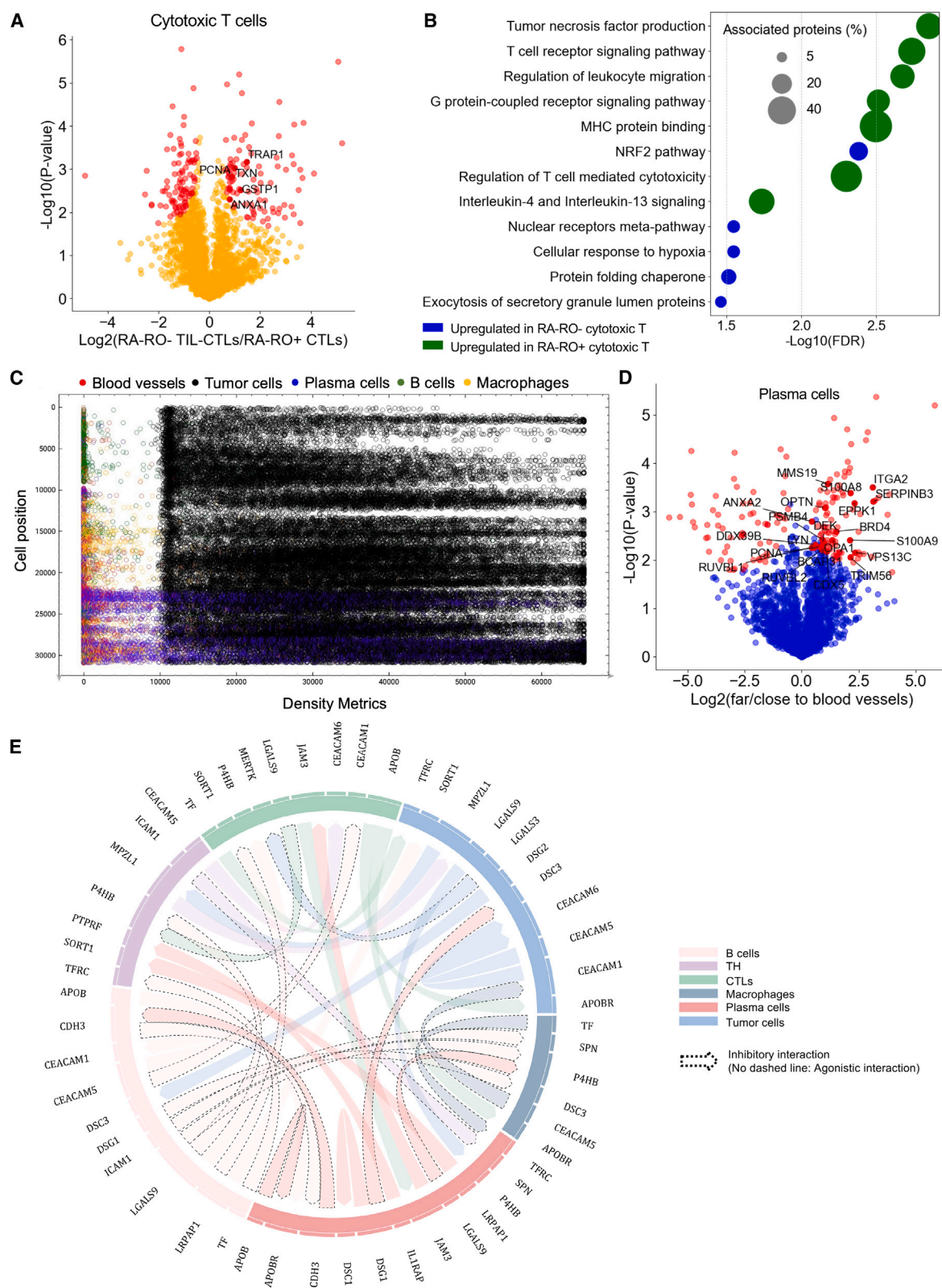
Following our analysis of how the spatial proximity to CTLs impacts the functional state of tumor cells, including their response to the cytotoxic activity of CTLs as judged by their proteomes, we next explored how CTLs themselves undergo proteomic alterations as a function of the TME. As described for the CRC case above, we first obtained proteomic profiles for the most abundant CTL subpopulations: CD45RA–CD45RO– (RA–RO–) CTLs and CD45RA–CD45RO+ (RA–RO+) CTLs (Figure 5C). We observed significant proteomic differences between RA–RO– CTLs residing in the tumor parenchyma (RA–RO– TIL-CTLs) and those in the stroma (RA–RO– stroma-CTLs), with 59 proteins differentially expressed (Figure S5F). RA–RO– TIL-CTLs exhibited downregulation of LAMA1, LAMA4, TNC, and ITGA6 compared with stroma-CTLs, suggest-

ing reduced adhesion to extracellular matrix components, potentially facilitating increased T cell mobility and improved penetration into the tumor parenchyma^{39,40} (Figure S5F). We identified 177 proteins that were significantly differentially regulated between RA–RO– TIL-CTLs and RA–RO+ CTLs, which were only present in the stroma (Figures 7A and F5D). RA–RO+ CTLs exhibited higher cytotoxic potential as judged by pathway analysis (Figure 7B). Additionally, RA–RO– TIL-CTLs adapted to the TME strategically, as indicated by upregulated response to hypoxia (including NRF2). Compared with their RA–RO+ counterparts residing in the stroma, this would facilitate their penetration into deeper tumor regions and more hypoxic tumor regions⁴¹ (Figures 7A and 7B). Consistent with this, molecules including ANXA1, GSTP1, PCNA, TRAP1, and TXN, which play crucial roles in hypoxia adaptation,^{42,43} were significantly upregulated in RA–RO– CTLs, with fold changes ranging between 1.5 and 4 (Figures 7A and S6A). To validate these proteomic findings, we conducted IF analysis of HIF-1 α to identify hypoxic tumor regions (Figure S6B). Notably, CD8+ CTLs residing in hypoxic tumor areas showed higher expression levels of ANXA1, GSTP1, PCNA, TRAP1, and TXN compared with those in normoxic regions, consistent with our proteomic findings (Figure S6C). This IF analysis cross-validates the MS-based proteomics data, confirming the adaptation of TIL-CTLs to hypoxia in the TME. We also compared the proteomes of RA–RO+ TH and RA–RO– TH and found no significant pathway differences between these subpopulations (Figure S6D).

In addition to the CTLs and macrophages investigated above, other cell types are involved in immune surveillance of tumor cells, including plasma cells, which are plentiful in the stroma (Figure 6E). To visualize the spatial distribution of these cell types, we plotted cellular density metrics along the x axis position, where a higher density metrics value indicates a greater density at that position. This visualization effectively distinguished blood vessels (with density metrics from 0 to 10,000) from tumor cells (with density metrics from 10,000 to 65,000), accurately representing their spatial density (Figure S6E). We extended our analysis to encompass tumor cells, blood vessels, B cells, and plasma cells, which revealed widespread distribution from vessel-rich areas to tumor sites (Figure 7C). Therefore, to explore proteomic differences in immune cells between vessel-rich areas and tumor sites, we isolated immune populations based on their proximity to blood vessels—those close to (<85 μm) and those far from (>85 μm) blood vessels, corresponding to being near and distant from tumor sites, respectively. We established a cutoff at 85 μm

Figure 6. Proteomic heterogeneity in tumor cells correlates with proximity to cytotoxic T cells (CTLs)

- Diagram illustrating the spatial localization of cells isolated from a tonsil cancer tissue for proteome profiling.
- Principal-component analysis (PCA) of tumor cell proteomes reveals distinct clustering based on proximity to CTLs in tumor parenchyma (TIL-CTLs).
- Heatmap highlighting significantly differentially expressed proteins among tumor cells with different proximity to TIL-CTLs (686 ANOVA significant proteins, FDR < 0.05, $n = 3$). Gene Ontology (GO) term analysis of indicated clusters is shown.
- Profile plots of key molecules in interferon, interleukin-12 signaling pathways, and phagocytic processes.
- Representative images illustrating tumor islands, blood vessels, and the distribution of B cells, plasma cells, and macrophages. The colored areas represent tumor or blood vessel densities, with a color gradient from red (highest density) to blue (lower density).
- Dot plot displaying the distance of tumor cells, TIL-CTLs, and macrophages to blood vessels. The white dotted line represents the median distance between tumor cells and blood vessels, the blue dotted line shows the median distance between TIL-CTLs and blood vessels, and the orange dotted line marks the median distance between macrophages and blood vessels.
- PCA of proteomes of tumor cells, lymphocytes, and macrophages.



(legend on next page)

because this was half of the median distance between tumor cells and blood vessels (Figure S6E).

Plasma cells located distant from vessels and closer to the tumor upregulated the stress responses, interleukin-3/5, and GM-CSF, along with increased chemotaxis response. This suggests active migration toward the tumor site and adaptation to the stress of the TME for effective immune surveillance. Conversely, plasma cells positioned near vessels and away from the tumor core upregulated pathways associated with RHO GTPase effectors, calcium ion response, and actin cytoskeletal organization, perhaps indicating strategic motility adaptation within the TME⁴⁴ (Figures 7D and S6F). Together, these findings highlight the dynamic nature of immune responses and the adaptive capabilities of CTLs and plasma cells in the TME, emphasizing the importance of spatial context in understanding the functional states of immune cells within the complex tumor ecosystem.

We conducted an analysis of cell-cell communication within the TME of a tonsil tumor, focusing on significant ligand-receptor interactions among CTLs, TH, macrophages, tumor cells, B cells, and plasma cells. Using the CellPhone DB dataset, we mapped key intercellular signaling interactions that underpin the immune landscape and tumor dynamics in this tonsil cancer case (Figure 7E).^{45,46} Our analysis revealed a highly interactive network between tumor cells and immune populations, indicative of active tumor-immune engagement. Tumor cells exhibited significant expression of CEACAM family members, which interacted with both immune cells—particularly CTLs and B cells—and among themselves. These interactions may enhance tumor recognition and destruction by CTLs.⁴⁷ We observed interactions between CTLs and tumor cells through TF and TFRC, and between tumor cells and plasma cells through LGALS9 and P4HB. Positive regulatory interactions were identified between TH cells and tumor cells or CTLs via MPZL1-MPZL1 and CEACAM5-CEACAM6. Metabolic crosstalk between tumor cells and CTLs or B cells was observed through APOB. The presence of both positive and negative interactions between immune and tumor cells indicates modulation of immune responses. For example, we identified interactions such as LGALS3 and MERTK between tumor cells and CTLs, ICAM1 and SPN between TH cells and plasma cells, and DSG1 and DSC3 or ICAM1 and SPN between B cells and macrophages. These interactions involve CEACAMs, GAL proteins, and apolipoprotein pathways. Our findings demonstrate active ligand-receptor dynamics in this immunologically hot tonsil tumor case, where CTLs and TH cells are interacting with tumor cells, involving both metabolic and regulatory inputs from surrounding immune and stromal cells.^{47,48}

DISCUSSION

In this study, we introduce highly multiplexed, imaging-powered Deep Visual Proteomics (mipDVP), which integrates spatial information from multiplexed imaging with the unbiased, deep protein discovery capabilities of DVP. This method overcomes limitations of existing technologies, such as the reliance on pre-defined antibody panels in imaging mass cytometry, CODEX, and digital spatial profiling, and the limited proteomics coverage of MALDI imaging.^{7,9} By capturing proteomic diversity at the single-cell-type level, mipDVP provides a more comprehensive view of signaling proteins that targeted approaches may overlook, enabling the unbiased exploration of the intricate cellular proteomic landscape.^{7,9} Unlike transcriptomics, which reveals the potential for gene expression, proteomics reflects the actual protein products that execute cellular functions and mediate intercellular communication. This distinction is critical because the abundance of proteins often does not directly correlate with mRNA levels due to various complex regulatory mechanisms.⁹ Although this study focuses on protein identification and quantification, future applications of spatial proteomics could extend to detecting post-translational modifications, which are key regulators of protein activity, localization, and interactions.⁴⁹

Applying mipDVP to CRC and a tonsil cancer samples revealed distinct proteomic signatures and spatial compartmentalization within their TMEs. In the CRC case, we identified compartmentalization of immunosuppressive macrophages and T cells, likely mediated by cell-cell adhesion, extracellular matrix secretion, or chemotactic gradients.^{28,50} These findings suggest potential therapeutic strategies, such as reprogramming macrophages or enhancing T cell infiltration and activity in immune-excluded tumors, particularly in mismatch-repair-proficient and microsatellite-stable cases.⁵⁰

In the tonsil cancer case, our 22-plex imaging revealed abundant immune cell infiltration, with proteomic analysis uncovering significant tumor cell heterogeneity influenced by proximity to tumor-infiltrating lymphocytes (TILs). Notably, CD45RA-CD45RO- (RA-RO-) TIL-CTLs upregulated hypoxia response pathways, facilitating their adaptation to hypoxic tumor regions. These findings underscore the importance of considering spatial context and cellular interactions when evaluating tumor immunological status and responsiveness to immunotherapies.

Our approach enabled mapping of intercellular communication networks through ligand-receptor interaction analysis. In the hot tonsil cancer case, both inhibitory and stimulatory

Figure 7. Cytotoxic T cells in tumor parenchyma (TIL-CTLs) exhibit strategic adaptations to hypoxic tumor microenvironment

(A) Volcano plot depicting pairwise proteomic comparisons between CD45RA-CD45RO- (RA-RO-) CTLs and RA-RO+ CTLs. Significantly enriched proteins are shown as red dots (two-sided t test, FDR < 0.05, S0 = 0.1, n = 3).

(B) Gene Ontology (GO) term analysis of significantly altered proteins in RA-RO- TIL-CTLs compared with their RA-RO+ counterparts, highlighting the up-regulation of pathways related to hypoxia response and NRF2 signaling.

(C) Density metrics displaying cell type densities relative to nuclear X position, with higher values signifying greater density.

(D) Volcano plot depicting pairwise proteomic comparisons between plasma cells in regions far from and close to blood vessels. Significantly enriched proteins are highlighted as red dots (two-sided t test, FDR < 0.05, S0 = 0.1, n = 3).

(E) Mapping immune dynamics through ligand-receptor interaction analysis. Chord diagrams illustrate intercellular communications via ligand-receptor interactions. Pairs with statistical significance ($p < 0.05$ and $|R| > 0.8$) were selected. Distinct color-coding represents different cell types, while directional arrows indicate the nature of the interactions: solid arrows denote positive regulatory signals ($R > 0.8$), and dashed arrows represent inhibitory signals ($R < -0.8$).

interactions involving CTLs and TH cells were observed, characterized by signaling molecules such as CEACAMs, LGALS, and metabolic regulators. These findings highlight the dynamic modulation of immune responses within the TME and suggest potential intervention points to enhance anti-tumor immunity.^{47,48}

MipDVP offers several advantages over existing methods. By combining spatial information with deep proteomic data, it enables the discovery of specific proteins and pathways, providing a detailed understanding of the interactions between tumor and immune cells. This approach could also optimize strategies for engineered immune cells, such as chimeric antigen receptor T (CAR-T) cells and T cell-receptor-engineered T (TCR-T) cells, by identifying their optimal spatial positioning and functional states within the TME.⁵¹ As this technology is refined and applied to diverse cancer types, it holds the potential to transform personalized oncology by tailoring treatment strategies and monitoring therapeutic responses based on the precise profiling of an individual patient's TME.

Limitations of the study

Although this study provides valuable insights into tumor-immune crosstalk and proteomic landscapes, a key limitation is the small number of cases analyzed. Larger cohorts encompassing diverse tumor types and microenvironmental conditions are needed to validate and generalize these findings. Additionally, although we employed robust computational frameworks and ligand-receptor interaction analyses, capturing dynamic signaling interactions from single-time point proteomic data remains inherently challenging. Despite these limitations, the ability to analyze diverse cell types within the same sample exemplifies the precision oncology paradigm, offering a strong foundation for future research.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Prof. Matthias Mann (mmann@biochem.mpg.de).

Materials availability

- This study did not generate new, unique reagents.

Data and code availability

- The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository and are publicly available under the accession number PXD057118.⁵² The multiplexed imaging data have been deposited in BioRxiv under accession number S-BIAD1116.⁵³
- This paper does not report any original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: X.Z., A.M., and M.M.; methodology: X.Z., A.M., and M.M.; data curation: X.Z.; investigation: X.Z.; validation: X.Z.; formal analysis: X.Z.; visualization: X.Z.; project administration: X.Z., A.M., and M.M.; supervision: M.M.; funding acquisition: M.M.; writing – original draft: X.Z. All authors contributed to the review and editing of the manuscript.

DECLARATION OF INTERESTS

M.M. is an indirect investor in Evosep Biosystems and OmicVision Biosciences. A.M. is a co-founder and chief scientific officer of OmicVision Biosciences.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT 4o and Anthropic Sonnet 3.5 to help edit the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-CD3-APC	Miltenyi Biotec	130-120-269; RRID:AB_2876933
anti-FOXP3-PE	Miltenyi Biotec	130-127-808; RRID:AB_2905207
anti-Mast Cell Tryptase-APC	Miltenyi Biotec	130-125-278; RRID:AB_2857771
anti-PD1-PE	Miltenyi Biotec	130-117-384; RRID:AB_2727929
anti-CD107a-APC	Miltenyi Biotec	130-126-203; RRID:AB_2889419
anti-CD38-PE	Miltenyi Biotec	130-126-438; RRID:AB_2904976
anti-HLA-DR-FITC	Miltenyi Biotec	130-123-076; RRID:AB_2857572
anti-CD4-PE	Miltenyi Biotec	130-127-906; RRID:AB_2921815
anti-CD45RA-APC	Miltenyi Biotec	130-112-097; RRID:AB_2819365
anti-CD20-FITC	Miltenyi Biotec	130-118-292; RRID:AB_2857425
anti-CD8-PE	Miltenyi Biotec	130-117-201; RRID:AB_2857415
anti-Ki67-FITC	Miltenyi Biotec	130-117-691; RRID:AB_2733585
anti-CD45RO-APC	Miltenyi Biotec	130-113-556; RRID:AB_2733381
anti-Cytokeratin-FITC	Miltenyi Biotec	130-112-743; RRID:AB_2651496
anti-CD57-PE	Miltenyi Biotec	130-111-810; RRID:AB_2658748
anti-CD11b-PE	Miltenyi Biotec	130-128-773; RRID:AB_2904691
anti-Podoplanin-APC	Miltenyi Biotec	130-126-165; RRID:AB_2905370
anti-CD68-PE	Miltenyi Biotec	130-128-345; RRID:AB_2905104
anti-Vimentin-FITC	Miltenyi Biotec	130-116-508; RRID:AB_2727581
anti-SMA-FITC	Miltenyi Biotec	130-123-363; RRID:AB_2857589
anti-CD31-PE	Miltenyi Biotec	130-128-769; RRID:AB_2904938
anti-ARG1-PE	Miltenyi Biotec	130-127-839; RRID:AB_2904631
anti-ANXA1-FITC	Miltenyi Biotec	130-119-352; RRID:AB_2857448
anti-PCNA-PE	Miltenyi Biotec	130-114-512; RRID:AB_2726680
anti-CLC	Abnova	ABNOPAB23850; RRID: AB_3271519
anti-EPX	Abcam	ab238506; RRID:AB_2923367
anti-CD8	Abcam	ab17147; RRID:AB_443686
anti-HIF1 α	Abcam	ab51608; RRID:AB_880418
anti-GSTP1	Proteintech	15902-1-AP; RRID:AB_2116216
anti-TRAP1	Proteintech	10325-1-AP; RRID:AB_2303284
anti-TXN	Proteintech	66475; RRID:AB_2881841
anti-HIF1 α -FITC	Novus	NB100-105F; RRID:AB_3148665
anti-CD4-FITC	R&D	FAB8165G; RRID:AB_2728839
Donkey anti-rabbit IgG Alexa Fluor 647	Thermo Fisher Scientific	A31573; RRID:AB_2536183
Donkey anti-mouse IgG Alexa Fluor 568	Thermo Fisher Scientific	A10037; RRID:AB_11180865
Donkey anti-rabbit IgG Alexa Fluor 488	Thermo Fisher Scientific	A21206; RRID:AB_2535792
Donkey anti-mouse IgG Alexa Fluor 647	Thermo Fisher Scientific	A31571; RRID:AB_162542
Donkey anti-rabbit IgG Alexa Fluor 568	Thermo Fisher Scientific	A10042; RRID:AB_2534017
Biological samples		
Formalin-fixed paraffin-embedded (FFPE) human tonsil	ProteoGenex	#68893B2(2)
FFPE human tonsil cancer	ProteoGenex	#652323T2(3)

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Vectabond	Vector Laboratories	SP-1800-7
Xylene	Carl Roth	4436.1
Ethanol	Merck	1070172511
EDTA buffer	Merck	E1161
Glycerol	Merck	G7757
TBS protein-free blocking buffer	LI-COR	927-80000
Staining wash buffer	Agilent	GC80711-2
DAPI	Abcam	ab228529
Anti-fade fluorescence mounting medium	Abcam	ab104135
Triethylammonium bicarbonate	Merck	T7408
n-dodecyl- β -D-maltoside	Merck	850520P
Acetonitrile	ThermoFisher	A955-1
LysC	Promega	VA1170
Trypsin	Promega	V5280
Trifluoroacetic acid	VWR	153112E
Formic	Merck	5.33002
Formaldehyde, CH ₂ O	Sigma-Aldrich	252549
Formaldehyde, CD ₂ O	Sigma-Aldrich	492620
Formaldehyde, ¹³ CD ₂ O	Sigma-Aldrich	596388
Sodium cyanoborohydride	Sigma-Aldrich	8180530025
Ammonia	Merck	5.33003
LC-MS-grade water	VWR	83645.290
Critical commercial assays		
MACSima stain support kit	Miltenyi Biotec	130-127-574
Deposited data		
Mass spectrometry proteomics data	This paper	PRIDE: PXD057118
MACSima imaging data	This paper	BiolImages: S-BIAD1116
Software and algorithms		
MACS iQ View Software v1.2.2	Kinkhabwala et al. ²¹	https://www.miltenyibiotec.com/DK-en/products/macs-iq-view-spatial-biology-software.html
Biology Image Analysis (BIAS) software v1.3.0	Mund et al. ¹⁵	https://single-cell-technologies.com/bias-2/
LMD software v8.3	Mund et al. ¹⁵	https://www.leica-microsystems.com/products/microscope-software/p/leica-lmd-software/
MSConvert software	Chambers et al. ⁵⁴	https://proteowizard.sourceforge.io/
DIA-NN v1.8.1	Demichev et al. ²⁶	https://github.com/vdemichev/DiaNN
FragPipe v18.0	Kong et al. ⁵⁵	https://fragpipe.nesvilab.org/
ClueGo v2.5.10	Bindea et al. ⁵⁶	https://apps.cytoscape.org/apps/cluego
CellPhoneDB v5.0.0	Troulé et al. ⁴⁵	https://www.cellphonedb.org/
Other		
Membrane PEN slides	Zeiss	415190-9041-000
Glass slides	Fisher Scientific	10149870
MACSima two-well imaging frame	Miltenyi Biotec	130-124-675
MACSima running buffer	Miltenyi Biotec	130-121-565
Thermal cycler	Bio-Rad	S1000
Aurora Elite TS analytical column	IonOpticks	AUR3-15075C18-TS
Aurora Elite CSI third-generation column	IonOpticks	AUR3-15075C18-CSI

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Tissue specimens

Formalin-fixed paraffin-embedded (FFPE) tissue specimens of a human tonsil and a tonsil cancer were obtained from ProteoGenex (Figure S1). The tonsil cancer specimen was from a 48-year-old Caucasian male diagnosed with grade G3, T2N2bM0, stage IV primary tonsil squamous cell carcinoma, with no prior chemotherapy, radiation, or hormonal treatment (sample #652323T2(3)). The normal tonsil specimen was from a 49-year-old Caucasian male (sample #68893B2(2)). FFPE tissue from a colorectal cancer (CRC) was sourced from Zealand University Hospital in Denmark and was fully anonymized. All specimens were acquired with informed consent, following standard protocols, and were approved by the relevant ethics committees, including those at ProteoGenex and the Health Research Ethics Approval (VEK).

METHOD DETAILS

Tissue section preparation

Membrane PEN slides (Zeiss 415190-9041-000) were exposed to UV light for 1 hour and coated with Vectabond reagent (Vector Laboratories SP-1800-7) for 5 minutes. FFPE tissue sections, cut at a thickness of 3 μ m, were then mounted on the treated PEN slides or conventional glass slides (Fisher Scientific 10149870), left to dry overnight at 37°C, and subsequently heated at 60°C for 20 minutes to enhance tissue adhesion. Paraffin was removed and the sections were rehydrated by sequentially passing them through xylene (Carl Roth 4436.1) and decreasing concentrations of ethanol (Merck 1070172511) as follows: xylene (3 $\times$ 5 minutes), 100% ethanol (2 $\times$ 2 minutes), 95% ethanol (2 minutes), 80% ethanol (2 minutes), 70% ethanol (2 minutes), and distilled deionized water (2 minutes). Antigen retrieval was performed using EDTA buffer (Merck E1161) with 10% glycerol (Merck G7757), heated at 88°C in a water bath for 45 minutes. For MACSima multiplexed staining, the two-well imaging frame (Miltenyi Biotec 130-124-675) was promptly mounted onto the slide, followed by the addition of 550 μ L of running buffer (Miltenyi Biotec 130-121-565).

MACSima multiplexed staining and imaging

The MACSima Imaging System by Miltenyi Biotec Biotec is a fully automated instrument seamlessly integrating liquid handling with widefield microscopy for cyclic immunofluorescence imaging.²¹ Before initiating the MACSima instrument, a DAPI pre-staining step was performed. The running buffer was removed from the sample, which was then incubated for 10 minutes with a 1:10 dilution of DAPI staining solution (Miltenyi Biotec 130-127-574). After staining, the sample underwent three washing steps with running buffer (Miltenyi Biotec 130-121-565), followed by the addition of 550 μ L of running buffer. The antibodies used in this study were obtained from Miltenyi Biotec, with all antibodies conjugated to FITC, PE, or APC fluorochromes. Staining conditions, including dilution factors, staining cycle sequences, and photobleaching steps, were optimized using normal tonsil tissue with unstained negative controls (Figures S1C and S2A). To minimize nonspecific binding, FCR block (Miltenyi Biotec 130-127-574) was used.

For a CRC membrane slide staining, the first step involved imaging and bleaching autofluorescence in the FITC, PE, and APC channels. Subsequent cycles included: First cycle: Incubation with anti-CD3-APC (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-120-269); Second cycle: Incubation with anti-HLA-DR-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-123-076) and anti-CD4-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-127-906); Third cycle: Incubation with anti-CD20-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-118-292) and anti-CD8-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-117-201); Fourth cycle: Incubation with anti-Ki67-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-117-691); Fifth cycle: Incubation with anti-Cytokeratin-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-112-743) and anti-CD11b-PE (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-128-773); Sixth cycle: Incubation with anti-Podoplanin-APC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-126-165), anti-CD68-PE (1:100, 10 minutes, room temperature, Miltenyi Biotec 130-128-345), and anti-Vimentin-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-116-508); Seventh cycle: Incubation with anti-SMA-FITC (1:100, 10 minutes, room temperature, Miltenyi Biotec 130-123-363) and anti-CD31-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-128-769).

For a tonsil cancer membrane slide staining, the first step involved imaging and bleaching autofluorescence in the FITC, PE, and APC channels. The subsequent cycles were as follows: First cycle: Incubation with anti-CD3-APC (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-120-269) and anti-FOXP3-PE (1:13, 10 minutes, room temperature, Miltenyi Biotec 130-127-808); Second cycle: Incubation with anti-Mast Cell Tryptase-APC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-125-278) and anti-PD1-PE (1:13, 10 minutes, room temperature, Miltenyi Biotec 130-117-384); Third cycle: Incubation with anti-CD107a-APC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-126-203) and anti-CD38-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-126-438); Fourth cycle: Incubation with anti-HLA-DR-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-123-076), anti-CD4-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-127-906), and anti-CD45RA-APC (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-112-097); Fifth cycle: Incubation with anti-CD20-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-118-292) and anti-CD8-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-117-201); Sixth cycle: Incubation with anti-Ki67-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-117-691), anti-CD57-PE (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-111-810), and anti-CD45RO-APC (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-113-556); Seventh cycle: Incubation with anti-Cytokeratin-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-112-743).

and anti-CD11b-PE (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-128-773); Eighth cycle: Incubation with anti-Podoplanin-APC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-126-165), anti-CD68-PE (1:100, 10 minutes, room temperature, Miltenyi Biotec 130-128-345), and anti-Vimentin-FITC (1:50, 10 minutes, room temperature, Miltenyi Biotec 130-116-508); Ninth cycle: Incubation with anti-SMA-FITC (1:100, 10 minutes, room temperature, Miltenyi Biotec 130-123-363) and anti-CD31-PE (1:25, 10 minutes, room temperature, Miltenyi Biotec 130-128-769).

Hybrid autofocusing was achieved using a combination of hardware autofocus and image-based autofocusing, optimized through the DAPI image. Imaging was performed with exposure series of 6, 36, and 216 ms in the APC channel; 5, 20, and 80 ms in the PE channel; and 8, 32, and 128 ms in the FITC channel. Photobleaching was employed to eliminate residual signals. The energy for photobleaching was 400J, 160J and 700J for FITC, PE and APC, respectively. Image acquisition was done using an epifluorescence setup with a long-working-distance 20 $\times$ objective (NA 0.45, objective slides 1.0 mm).

Low-plex staining and imaging

Antigen retrieval was conducted by immersing the tissue sections on glass slides in EDTA buffer (Merck E1161) at 90°C for 30 minutes. Following this, the sections were blocked with TBS protein-free blocking buffer (LI-COR 927-80000) for 20 minutes at room temperature.

For CD4 + ANXA1 staining, the sections were incubated overnight at 4°C with primary antibodies (anti-CD4-PE, 1:100, Miltenyi Biotec 130-127-906; anti-ANXA1-FITC, 1:200, Miltenyi Biotec 130-119-352) in a humidified chamber. After three washes with 1 $\times$ wash buffer, each for 5 minutes, the sections were counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) before being examined under a Zeiss AxioScan7 microscope.

For CD4 + ARG1 staining, the sections were incubated overnight at 4°C with primary antibodies (anti-CD4-FITC, 1:100, R&D FAB8165G; anti-ARG1-PE, 1:200, Miltenyi Biotec 130-127-839) in a humidified chamber. After three washes with 1 $\times$ wash buffer (Agilent GC80711-2), each for 5 minutes, the sections were counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) before being examined under a Zeiss AxioScan7 microscope.

For CD4 + CLC/EPX staining, sections were incubated overnight at 4°C with primary antibodies (anti-CLC, 1:300, Abnova ABNOPAB23850; or anti-EPX, 1:300, Abcam ab238506) in a humidified chamber. After three 5-minute washes, sections were incubated with secondary antibody (Donkey anti-rabbit IgG Alexa Fluor 647, 1:500, Thermo Fisher Scientific A31573) in blocking buffer for 1 hour at room temperature in a dark chamber. Following three more 5-minute washes, the sections were incubated overnight with primary antibodies (anti-CD4-PE, 1:100, Miltenyi Biotec 130-127-906) at 4°C. The sections were then washed, counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature, and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) before imaging.

For CD8 + HIF1 α + ANXA1 staining, sections were incubated overnight at 4°C with primary antibody (anti-HIF1 α , 1:500, Abcam ab51608) in a humidified chamber. Following three washes in 1 $\times$ wash buffer for 5 minutes each, sections were incubated with secondary antibodies (Donkey anti-rabbit IgG Alexa Fluor 647, 1:500, Thermo Fisher Scientific A31573) in blocking buffer for 1 hour at room temperature in a dark chamber. After three additional washes for 5 minutes each, the sections were incubated overnight with primary antibody (anti-CD8-PE, 1:100, Miltenyi Biotec 130-117-201; anti-ANXA1-FITC, 1:200, Miltenyi Biotec 130-119-352) at 4°C. The sections were then washed, counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature, and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) for examination under a microscope.

For CD8 + HIF1 α + PCNA staining, sections were incubated overnight at 4°C with primary antibodies (anti-CD8, 1:200, Abcam ab17147; anti-HIF1 α , 1:500, Abcam ab51608) in a humidified chamber. After three 5-minute washes in 1 $\times$ wash buffer, the sections were incubated with secondary antibodies (Donkey anti-rabbit IgG Alexa Fluor 488, 1:500, Thermo Fisher Scientific A21206; Donkey anti-mouse IgG Alexa Fluor 647, 1:500, Thermo Fisher Scientific A31571) in blocking buffer for 1 hour at room temperature in a dark chamber. After three 5-minute washes, the sections were incubated overnight with primary antibody (anti-PCNA-PE, 1:200, Miltenyi Biotec 130-114-512) at 4°C. The sections were then washed, counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature, and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) for imaging under the Zeiss AxioScan7 microscope.

For CD8 + HIF1 α + GSTP1/TRAP1 staining, sections were incubated overnight at 4°C with primary antibodies (anti-GSTP1, 1:100, Proteintech 15902-1-AP; or anti-TRAP1, 1:100, Proteintech 10325-1-AP) in a humidified chamber. After three 5-minute washes, the sections were incubated with secondary antibody (Donkey anti-rabbit IgG Alexa Fluor 647, 1:500, Thermo Fisher Scientific A31573) in blocking buffer for 1 hour at room temperature in a dark chamber. Following three more 5-minute washes, the sections were incubated overnight with primary antibodies (anti-CD8-PE, 1:100, Miltenyi Biotec 130-117-201; anti-HIF1 α -FITC, 1:500, Novus NB100-105F) at 4°C. The sections were then washed, counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature, and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) before examination under the Zeiss AxioScan7 microscope.

For CD8 + HIF1 α + TXN staining, sections were incubated overnight at 4°C with primary antibodies (anti-TXN, 1:500, Proteintech 66475; anti-HIF1 α , 1:500, Abcam ab51608) in a humidified chamber. After three 5-minute washes in 1 $\times$ wash buffer, the sections were incubated with secondary antibodies (Donkey anti-rabbit IgG Alexa Fluor 488, 1:500, Thermo Fisher Scientific A21206; Donkey

anti-mouse IgG Alexa Fluor 647, 1:500, Thermo Fisher Scientific A31571) in blocking buffer for 1 hour at room temperature in a dark chamber. After three 5-minute washes, the sections were incubated overnight with primary antibody (anti-CD8-PE, 1:100, Miltenyi Biotec 130-117-201) at 4°C. The sections were then washed, counterstained with DAPI (1:1000, Abcam ab228529) for 5 minutes at room temperature, and mounted with anti-fade fluorescence mounting medium (Abcam ab104135) for imaging under the Zeiss AxioScan7 microscope.

MACSima multiplexed image analysis

For the analysis of MACSima multiplexed images, image preprocessing was performed using MACS iQ View Software (v1.2.2).²¹ This process included correcting image distortion, stitching, spatially aligning images through registration, and subtracting residual intensities from autofluorescence or incomplete erasure of previous cycle staining. During the preprocessing setup, the “Nikon 20× Plan Apochromat” lens was selected, with exposure set to “Automatic” at 100% scaling, and the “crop to DAPI” option enabled. Image segmentation was conducted using nuclei and cell membrane markers: nucleus segmentation utilized the “Advanced Morphology for Tissue” method with DAPI as the reference channel, while cytoplasm segmentation employed the “Constrained Donut” method with automatic constraint channels.

Cellular classification was conducted using a gating-based strategy, identifying 21 distinct cell types and subtypes. These included Cytokeratin+ tumor cells; CD3+CD8+ cytotoxic T cells, which were further classified into subtypes including CD3+CD8+CD45RA-CD45RO-, CD3+CD8+CD45RA+CD45RO+, CD3+CD8+CD45RA+CD45RO-, and CD3+CD4+CD45RA-CD45RO+; CD3+CD4+ helper T cells, subdivided into CD3+CD4+CD45RA-CD45RO-, CD3+CD4+CD45RA+CD45RO+, CD3+CD4+CD45RA+CD45RO-, and CD3+CD4+CD45RA-CD45RO+ subtypes; CD20+CD3-CD38- B cells, with subtypes CD20+CD3-CD38-CD45RA+CD45RO+, CD20+CD3-CD38-CD45RA-CD45RO-, CD20+CD3-CD38-CD45RA+CD45RO-, and CD20+CD3-CD38-CD45RA-CD45RO+; as well as CD38+CD20-Ki67- plasma cells, CD57+CD107a+HLA-DR+ NK cells, CD68+HLA-DR+ macrophages, Mast Cell Tryptase+ mast cells, Vimentin+SMA+ fibroblasts, CD11b+CD68- granulocytes, Podoplanin+ lymphatic endothelial cells, and CD31+ blood vessel endothelial cells (Figure S2C).

Based on the expression of CK, CD3, CD4, CD8, CD20, CD38, CD57, CD11b, CD68, CD31, Podoplanin, HLA-DR and Ki67, Uniform manifold approximation and projection (UMAP) was applied to reveal four main distinct cellular clusters, including tumor cells, lymphocytes, myeloid cells and endothelial cells. The lymphocyte cluster was further subdivided into B cells, cytotoxic T cells, helper T cells, plasma cells, and NK cells. Spatial distance analysis was performed using Euclidean distance within MACS iQ View Software. This involved creating distance metrics for a classified cell type (e.g., cytotoxic T cells) in the “Workflow Editor,” then plotting histograms in “Feature Plot” to visualize the distribution of tumor cells at various distances (e.g., close proximity <30 μm, intermediate distances >30 μm and <60 μm, and further away >60 μm and <90 μm) from cytotoxic T cells. Distance heatmaps were generated by calculating multiple cell distance metrics, which were subsequently arcsinh normalized and averaged. Specifically, the pixel distances are measured from the center of each cell to the nearest neighboring cell of a specified type. For each cell of type A, the distance to its closest neighbor of type B is computed. The arcsinh transformation is then applied to these distances, which helps stabilize variance across a broad range of values, preserving both small and large distance information.

To align the masks of cells of interest with the membrane section for downstream laser microdissection, the masks were exported from MACS iQ View Software and imported into Biology Image Analysis (BIAS) software (v1.3.0).¹⁵ Three reference points were set up, and the corresponding mask coordinates were exported as XML files from BIAS. These files were then imported into Leica LMD software (v8.3) for precise alignment and microdissection.

4-plexed image analysis

Images from the Zeiss AxioScan7 microscope were imported into BIAS for deep learning-based cell segmentation, machine learning-based classification, and subsequent single-shape export for laser microdissection. For cell segmentation, a pre-trained deep neural network model of generic cytoplasm segmentation was used. The detection and contour confidence levels were both set to 50%. Cells with areas smaller than 5 μm² or larger than 200 μm² were excluded from analysis. To accurately classify T cells, a BIAS-integrated multilayer perceptron (MLP) feedforward neural network was trained on manually identified T cells, with a weight scale and momentum parameter set to 0.01 and the number of iterations set to 10,000. The classification model was evaluated using 10-fold cross-validation. Prior to classification, regions of interest were annotated, and a marker expression feature table was exported for quantification of marker expression.

Laser microdissection

To align the masks of the cells of interest with the membrane section for downstream laser microdissection, the masks were exported from MACS iQ View Software and imported into Biology Image Analysis (BIAS) software (v1.3.0). Three reference points were established, and the corresponding mask coordinates were exported as XML files from BIAS using the “greedy” sorting method. These XML files were then imported into Leica LMD software (v8.3) for precise alignment and microdissection. After aligning the reference point and mask coordinates to the Leica LMD7 laser microdissector at 63× magnification, the laser microdissection of the cells of interest was performed under the following parameters: middle pulse cutting method was selected, laser power was set to 35, aperture was set to one, cutting speed was set to 25, middle pulse count was set to four, final pulse was set to one, head current was at 50%, and pulse frequency was set to 2,600 Hz. High cutting precision was achieved using an HC PL FLUOTAR L ×63/0.70 objective.

The Leica Laser Microdissection software (v8.3) was utilized for fully automated excision and collection of contours. Each cell type of interest was excised in triplicates or quadruplicates from the same membrane section, with each section covering a surface area of $37,000 \mu\text{m}^2$ (around 200 lymphocytes or 100 macrophages). The cells were collected into 384-well plates, deliberately omitting the outermost rows and columns. Following microdissection, the collected cells were sedimented at the base of each well through centrifugation at 2,000 g for 10 minutes, assisted by 100% acetonitrile (Thermo Fisher Scientific A955-1), followed by vacuum evaporation until completely dried.

MS sample preparation

Lysis was performed using 4 μL of 60 mM triethylammonium bicarbonate (TEAB, Merck T7408) and 0.01% n-dodecyl- β -D-maltoside (DDM, Merck 850520P), followed by incubation at 95°C for 60 minutes in a thermal cycler (Bio-Rad S1000 with 384-well reaction module). To this, 1 μL of 60% acetonitrile (Thermo Fisher Scientific A955-1) was added, and the samples were heated at 75°C for 1 hour. Protein digestion was achieved using 4 ng LysC (Promega VA1170) and 6 ng trypsin (Promega V5280) overnight at 37°C .

The CRC-derived samples were label-free and directly subjected to Liquid chromatography with tandem mass spectrometry (LC-MS/MS) analysis after protein digestion. The samples were acidified with 1.22 μL of 10% Trifluoroacetic acid (TFA, final concentration 1%, VWR 153112E), vacuum dried, and stored at -20°C or resuspended in Evosep buffer A (0.1% formic acid, Merck 5.33002) for direct Evtip loading.

For samples derived from the tonsil cancer tissue, dimethyl labeling was performed following the method described by Thielert and Itang et al.¹⁶ A parallel bulk tissue slice was digested as the reference channel. The labeling involved sequential addition of 1 μL of 1.5% formaldehyde (CH_2O , Sigma-Aldrich 252549; CD_2O , Sigma-Aldrich 492620; or $^{13}\text{CD}_2\text{O}$, Sigma-Aldrich 596388) and 1 μL of 230 mM sodium cyanoborohydride (Sigma-Aldrich 8180530025), followed by incubation at room temperature for 1 hour on a bench-top mixer. To halt the reaction, 1 μL of 1.43% ammonia (Merck 5.33003) was added. The samples were acidified with 1.22 μL of 10% TFA (final concentration 1%), vacuum dried, and stored at -20°C or resuspended in Evosep buffer A (0.1% formic acid) for direct Evtip loading. Evtip loading involved using 25 ng of the bulk tissue as the reference channel ($\Delta 0$), combined with the target $\Delta 4$ and $\Delta 8$ channels.¹⁶

LC-MS/MS

The CRC-derived samples were analyzed using the Orbitrap Astral mass spectrometer (Thermo Fisher Scientific) interfaced with the Evosep One LC system (EvoSep). The Whisper40 SPD (samples per day) method with a gradient flow rate of 100 nL/min was employed, utilizing the Aurora Elite TS analytical column (IonOpticks AUR3-15075C18-TS) and the EASY-Spray™ source (Thermo Fisher Scientific). Mobile phases consisted of 0.1% formic acid (Merck 5.33002) in LC-MS-grade water (VWR 83645.290) as buffer A and 99.9% acetonitrile (Thermo Fisher Scientific A955-1)/0.1% formic acid (Merck 5.33002) as buffer B. All samples were recorded in DIA mode. The Orbitrap Astral mass spectrometer was utilized for full MS analyses with a resolution setting of 240,000 within a full scan range of 380–980 m/z. The automatic gain control (AGC) for the full MS was adjusted to 500%. The MS/MS isolation window was set to 8 Th, the maximum ion injection time to 14 ms, and the MS/MS scanning range between 380 – 980 m/z. Selected ions were fragmented by higher-energy collisional dissociation at a normalized collision energy (NCE) of 25%.

Samples derived from the tonsil cancer tissue underwent analysis using the Evosep One LC system (EvoSep) coupled to a timsTOF SCP mass spectrometer (Bruker). The Whisper40 SPD method was applied, employing the Aurora Elite CSI third-generation column (IonOpticks AUR3-15075C18-CSI) at 50°C within the Captive spray source (Bruker). Mobile phases consisted of 0.1% formic acid (Merck 5.33002) in LC-MS-grade water (VWR 83645.290) as buffer A and 99.9% acetonitrile (Thermo Fisher Scientific A955-1)/0.1% formic acid (Merck 5.33002) as buffer B. The timsTOF SCP was operated with an optimized dia-PASEF method, encompassing 8 dia-PASEF scans with variable width and 2 ion mobility windows per dia-PASEF scan.^{57,58} This spanned an m/z range from 300 to 1200 and an ion mobility range from 0.7 to 1.3 V/cm²/s. The mass spectrometer functioned in high sensitivity mode, with an accumulation and ramp time set at 100 ms, a capillary voltage of 1400 V, and collision energy following a linear ramp from 20 eV at $1/K_0 = 0.6 \text{ V/cm}^2/\text{s}$ to 59 eV at $1/K_0 = 1.6 \text{ V/cm}^2/\text{s}$.

Evaluation of Instrument Influence: To assess the performance of timsTOF and Astral, we collected bulk samples from different tumor microenvironment regions within the same cell population. Specifically, immune and stromal cells from both the lamina propria and tumor epithelial regions, as well as tumor epithelial cells from a CRC case, were collected in five technical replicates, each covering a surface area of $70,000 \mu\text{m}^2$. We used label-free quantification. In the lamina propria immune and stromal cell samples, timsTOF detected approximately 9,000 precursors, while Astral identified around 21,000. For immune and stromal cells from the tumor epithelial region, timsTOF detected about 15,000 precursors compared to Astral's 32,000. In the tumor epithelial cell samples, timsTOF identified around 17,000 precursors, whereas Astral detected about 42,000. Overall, Astral identified 2.3 times more precursors than timsTOF (Figure S7A). Regarding protein groups, timsTOF detected around 2,000, 2,800, and 3,800 in the lamina propria immune and stromal cells, tumor epithelial immune and stromal cells, and tumor epithelial cells, respectively. In comparison, Astral identified approximately 3,400, 4,400, and 5,800 protein groups in these samples. Across all three conditions, timsTOF detected over 6,000 protein groups with a median coefficient of variation (CV) of around 17%, while Astral identified more than 14,000 protein groups with a median CV of approximately 16% (Figures S7B and S7C). Overall, Astral detected 1.6 times more protein groups than timsTOF while maintaining a similar median CV. The overlap of protein groups detected by both methods was high: 86% in

lamina propria immune and stromal cells, 81% in tumor epithelial immune and stromal cells, and 80% in tumor epithelial cell samples (Figure S7D). This indicates a strong concordance between timsTOF and Astral in the protein groups identified.

MS data analysis

For label-free CRC-derived samples, the raw files were initially converted to the mzML file format using MSConvert software (Proteowizard),⁵⁴ with default parameters and 'Peak Picking' selected as the filter. Subsequently, mzML files were quantified in DIA-NN v1.8.1 against a human FASTA reference file (2023, UP000005640_9606) from the UniProt database, employing a direct-DIA approach.²⁶ DIA-NN search parameters included: Enzyme specificity set to 'Trypsin/P' with a maximum of one missed cleavage. Parameters for post-translational modifications like N-terminal methionine excision, methionine oxidation, and N-terminal acetylation were activated, allowing for a maximum of two variable modifications. Precursor FDR was limited to 1%, and both mass and MS1 accuracy were set to 15 p.p.m. Protein inference was conducted using 'Genes', with the neural network classifier in 'Single-pass mode' and quantification strategy as 'Robust LC (high precision)'. Cross-run normalization was set as 'RT-dependent', while library generation involved 'IDs, RT and IM Profiling'. Mass accuracy and MS1 accuracy were automatically inferred from the first run, with additional settings such as 'Use isotopologues', 'Match between runs (MBR)', 'Heuristic protein inference', and 'No shared spectra' enabled for optimal results.

For dimethyl-labeled tonsil cancer samples, the spectral library was constructed using five dda-PASEF single shots from a 50 ng bulk reference peptide. Spectra were searched with FragPipe v18.0 using MSFragger v3.5, Philosopher v4.4.0, and EasyPQP v0.1.32 against a human FASTA reference file (2023, UP000005640_9606).⁵⁵ The DIA_SpecLib_Quant workflow was applied with standard settings, except for fixed modifications where N-terminal and lysine mass shifts of 28.0313 Da were set, and methionine oxidation was defined as a variable modification. Carbamidomethylation was unselected due to samples not undergoing reduction and alkylation. One missed cleavage was permitted. The precursor charge ranged from 2 to 4, the peptide mass range was set from 300 to 1,800, and peptide length ranged from 7 to 30. The resulting library file was utilized in DIA-NN v1.8.1 to analyze the raw files from diaPASEF measurements.²⁶ DIA-NN search parameters mirrored those used for label-free samples, with similar settings for protein inference, quantification strategy, and additional optimizations. Additional commands were entered into the DIA-NN command line GUI, including (1) {--fixed-mod Dimethyl, 28.0313, nK}, (2) {--channels Dimethyl, 0, nK, 0:0; Dimethyl, 4, nK, 4.0251:4.0251; Dimethyl, 8, nK, 8.0444:8.0444}, (3) {--original-mods}, (4) {--peak-translation}, (5) {--ms1-isotope-quant}, (6) {--report-lib-info}, and (7) {--mass-acc-quant 10.0}. The processed DIA-NN output report table underwent additional analysis using the 'RefQuant' package implemented in Python. The RefQuant output was filtered for 'Lib.PG.Q.Value' < 0.01, 'Q.value' < 0.01 and 'Channel.Q.Value' < 0.15.¹⁶ Subsequently, the data was consolidated into protein groups using the MaxLFQ algorithm, implemented in the R package 'iq'.⁵⁹

Quantification and statistical analysis

The proteome datasets underwent filtering to include samples with at least 70% valid values, excluding proteins with more than 30% missing values for subsequent statistical analyses using Perseus.⁶⁰ Remaining missing values were imputed using the K-Nearest Neighbors (KNN) imputation method.⁶¹ All samples' intensity values underwent a logarithmic transformation ($\log_2$) and were used for principal component analysis (PCA) and identification of differentially expressed proteins.

Protein visualization employed volcano plots and heat maps, analyzed using readily available code in Python and R. For multi-sample comparisons (ANOVA) or pairwise proteomic analyses (two-sided unpaired t-test), a protein was deemed significantly differentially abundant if the FDR-adjusted *P*-value was less than 0.05.

Ontology enrichment analysis on the single-cell type proteomics dataset was conducted using ClueGo (v2.5.10), a Cytoscape (v3.1.1) plug-in, with default settings. We used the identified proteins to create customized reference sets comprising around 5000 unique genes to perform Fisher's exact tests. Term significance was corrected using the Benjamini-Hochberg method with an FDR threshold of less than 1%. The analysis encompassed the activation of both Gene Ontology term fusion and grouping.

To systematically interrogate intercellular communication at the proteomic level, we leveraged a computational framework incorporating the curated ligand-receptor interaction dataset from CellPhoneDB.⁴⁵ This allowed for an in-depth resolution of cell-cell signaling networks. We initiated the analysis by mapping known ligand-receptor pairs across distinct cellular populations, using an unbiased query of their expression levels in our dataset. Subsequently, pairwise correlation analyses between ligands and their corresponding receptors were conducted across heterogeneous cell types, employing Spearman's rank-order correlation to capture non-linear dependencies inherent in proteomic data. Correlation coefficients (*R* values) and their associated *P*-values were calculated. To ensure robustness, we isolated pairs that displayed statistical significance ($P < 0.05$ and $|R| > 0.8$) for further analysis, thereby excluding any low-confidence interactions. For visualization, we implemented a chord diagram representation.⁴⁶ The diagram not only captured the directional flow of signaling from ligand-expressing to receptor-expressing cells but also integrated multiple layers of information into its structure. Cell types were represented with distinct color-coding, and the directional arrows connecting ligands to receptors were meticulously encoded with dual significance: solid arrows denoted positive regulatory signals ($R > 0.8$), while dashed arrows represented inhibitory signaling ($R < -0.8$). Additionally, the width of each arrow was meticulously scaled to reflect the magnitude of the absolute *R* value, ensuring that the strength of each interaction was accurately conveyed.